34. Hypercomplex Quantities and Representation Theory

Continuation of §4. Direct products and direct intersections

A group D is called the **direct intersection** of two groups R and S with respect to G if

a) R and S are normal subgroups of G,

b) $R \cap S = D$,

c) $RS = G$.

Similarly, D is called the direct intersection of n groups $S_1, \dots, S_n$ if, after putting

$$R_i = S_1 \cap \cdots \cap S_{i-1} \cap S_{i+1} \cap \cdots \cap S_n,$$

one has

$$D = R_i \cap S_i$$

directly for each i.

It follows at once from the definition that D must be a normal subgroup of G.

If

$$D = S_1 \cap \cdots \cap S_n$$

is direct and, under the homomorphism $G \to G/D$, the groups G, S_i, D go over into $\overline{G}, \overline{S}_i, E$, then

$$E = \overline{S}_1 \cap \cdots \cap \overline{S}_n$$

is direct. Conversely, from every such representation

$$E = \overline{S}_1 \cap \cdots \cap \overline{S}_n$$

one returns to a direct representation

$$D = S_1 \cap \cdots \cap S_n.$$

This correspondence reduces theorems on direct intersections with arbitrary D to the special case $D = E$.

For $D = E$ and two factors, the conditions for direct product and direct intersection coincide. For n factors there is the following one-to-one relation between direct product and direct intersection.

I. If

$$G = A_1 \times \cdots \times A_n = A_i \times B_i,$$

then

$$E = B_1 \cap \cdots \cap B_n = B_i \cap C_i$$

directly, and $C_i = A_i$.

II. If

$$E = S_1 \cap \cdots \cap S_n = R_i \cap C_i$$

directly, then

$$G = R_1 \times \cdots \times R_n = R_i \times T_i,$$

and $T_i = S_i$.

For I, write

$$B = B_1 \cap \cdots \cap B_n = B_i \cap C_i.$$

Then $C_i \supseteq A_i$; we show that $C_i \subseteq A_i$. If, for example, $c \in C_1$, then c belongs to $B_2, \ldots, B_n$, and its component representation with respect to the factors A_i gives

$$c = a_1 a_2 \cdots a_n = a_1 e a_3 \cdots a_n = a_1 a_2 e \cdots a_n = \cdots = a_1 \cdots a_{n-1} e.$$

Since the product of all the A_i is direct, these representations agree. At every position except the first there appears once the identity element. Hence $c = a_1$, and so $C_1 \subseteq A_1$. Thus $C_i = A_i$ for all i. It follows that

$$B = B_i \cap A_i = E, \qquad B_i A_i = G,$$

so that the intersection is direct.

For II, write

$$H = R_1 \cdots R_n = R_i T_i.$$

Then $T_i \subseteq S_i$; we show that $S_i \subseteq T_i$. Let, for instance, $s \in S_1$. Since $G = S_1 \times R_1$, one may write

$$s = s_1 e = s_2 r_2 = \cdots = s_n r_n.$$

Put

$$t_1 = e r_2 \cdots r_n = t_i r_i.$$

Using the elementwise commutativity of R_i and S_i one obtains

$$s t_1^{-1} = s_i t_i^{-1},$$

an element of all the S_i, hence the identity. Thus $S_1 \subseteq T_1$, and so $T_1 = S_1$; similarly $T_i = S_i$ for every i. Consequently

$$H = R_i T_i = R_i S_i = G, \qquad R_i \cap T_i = R_i \cap S_i = E,$$

and therefore G is the direct product of the R_i.

§5. Completely reducible groups

A group is called **completely reducible** if it is the direct product of finitely many simple groups:

$$G = A_1 \times \cdots \times A_n.$$

In this case the groups

$$G, A_2 \times \cdots \times A_n, A_3 \times \cdots \times A_n, \ldots, A_n, E$$

form a composition series. Its length is n, and its composition factors are

$$A_1, A_2, \ldots, A_n.$$

If G is completely reducible, then every normal subgroup is a direct factor. The complementary factor can be chosen as the product of simple groups occurring in a prescribed product decomposition of G. The normal subgroup itself is completely reducible.

Indeed, suppose

$$G = HA_1A_2 \cdots A_n.$$

Put $H_0 = H$ and $H_i = H_{i-1}A_i$. Either $A_i \subseteq H_{i-1}$, in which case the factor A_i may be omitted, or $H_{i-1} \cap A_i$ is a proper normal subgroup of A_i and hence equals E; in that case

$$H_i = H_{i-1} \times A_i.$$

After omitting the superfluous factors the decomposition becomes

$$G = H \times A_{i_1} \times \cdots \times A_{i_s},$$

so H is a direct factor. Moreover

$$H \simeq G/(A_{i_1} \times \cdots \times A_{i_s}),$$

which exhibits H as completely reducible.

It follows that every decomposition of a completely reducible group into directly indecomposable factors is already a decomposition into simple factors. It therefore gives a composition series, and the factors are uniquely determined, up to isomorphism and order.

§6. Modules relative to a field. Hypercomplex systems

An almost trivial example of completely reducible groups with operators is furnished by modules with finite basis relative to a field, not necessarily commutative.

Let G be a right K-module, and assume that the unit element e of K is at the same time the identity operator: $ae = a$ for $a \in G$. Every submodule aK generated by a non-zero element a is simple; indeed it is equal to the module generated by any one of its non-zero elements. Thus, if H is the submodule generated by certain elements and if $a \notin H$, then

$$H \cap aK = 0,$$

and $H + aK$ is direct. Starting with a basis element a_1 one may adjoin further basis elements $a_2, a_3, \ldots$, obtaining

$$G = a_1K + a_2K + \cdots + a_nK$$

as a direct sum. Hence G is completely reducible.

The number n, the length of the composition series, is called the **rank** of G relative to K. Because the representation of the elements is unique, and because e is assumed to be the identity operator, the basis $a_1, \ldots, a_n$ is linearly independent.

Every submodule H is a direct summand:

$$G = H + R = (a_1K + \cdots + a_rK) + (a_{r+1}K + \cdots + a_nK).$$

Equivalently, every linearly independent basis of H can be extended to a linearly independent basis of G. The complementary basis elements may even be chosen from the original basis elements.

If $(a_1, \ldots, a_n)$ and $(c_1, \ldots, c_n)$ are two linearly independent bases, then

$$c_j = \sum_i a_i p_{ij},$$

or, in matrix notation,

$$(c_1, \ldots, c_n) = (a_1, \ldots, a_n)P.$$

Conversely,

$$(a_1, \ldots, a_n) = (c_1, \ldots, c_n)Q.$$

It follows that

$$(c_1, \ldots, c_n) = (c_1, \ldots, c_n)QP,$$

and, since the c_i are linearly independent,

$$QP = E.$$

The corresponding relation gives $PQ = E$ as well.

Special K-modules which are at the same time rings are the **hypercomplex systems**.

A ring $\mathfrak{o}$ which is also a right module relative to a commutative field K is called a hypercomplex system relative to K (also called, in the literature, an algebra over K) if the following hold.

1. The rank is finite; let $u_1, \ldots, u_n$ be a linearly independent basis.
2. Elements of K commute with multiplication in $\mathfrak{o}$:

$$(ab)x = a(bx) = (ax)b.$$

   One says that K is connected commutatively with $\mathfrak{o}$.
3. The unit element e of K is the identity operator:

$$ae = a \qquad (a \in \mathfrak{o}).$$

Because

$$\left(\sum_i u_i\alpha_i\right)\left(\sum_j u_j\beta_j\right) = \sum_{i,j}(u_iu_j)(\alpha_i\beta_j),$$

the system is completely determined, besides K, by the multiplication table

$$u_iu_j = \sum_k u_k\gamma_{ij}^k.$$

The coefficients γ_{ij}^k have to satisfy the familiar relations imposed by associativity.

If, as will often be assumed, $\mathfrak{o}$ has a unit element 1 with $1a = a1 = a$ for all a, then the elements $x \in K$ may be identified with $1x$. Thus K may be regarded as a subfield of $\mathfrak{o}$. In that case the hypercomplex system may be described as a ring of finite rank over a field lying in the center.

One example is the group ring of a finite group: take the group elements as basis elements u_i, and define multiplication by the group multiplication. The field K is arbitrary.

§7. Matrices

The square matrices (x_{ij}) with entries in a ring $\mathfrak{o}$ form a ring under the usual matrix multiplication and addition.

A matrix with entries in a field K is called **regular** if it has both a right inverse and a left inverse.

In §6 we saw that the transition matrix between two linearly independent bases of a K-module is regular. For regularity it is enough to have a right inverse. To see this, form a right K-module with linearly independent basis $(a_1, \ldots, a_n)$, and put

$$c_j = \sum_i a_ip_{ij}.$$

If P has a right inverse, then the a_i may be expressed through the c_j; hence the c_j are again a basis, and in particular linearly independent. Thus P is regular. The right inverse is then also the left inverse. The same argument applies if a left inverse is given first. If P has a left inverse, P is not a left zero divisor: from $PA = 0$ one obtains $A = P^{-1}PA = 0$.

Let P_{ij} denote, as usual, the matrix whose (i, j) entry is x, all other entries being zero; and let E_{ij} denote the matrix which has the unit element at position (i, j) and zeros elsewhere. Then

$$(x_{ij}) = \sum E_{ij} x_{ij}.$$

Thus the matrices over K form a K-module of rank n^2. The matrix units satisfy

$$E_{ij}E_{jk} = E_{ik}, \qquad E_{ij}E_{hk} = 0 \quad (j \neq h).$$

Consequently the matrices over K form a finite K-module and, when K is commutative, a hypercomplex system.

Chapter II. Non-commutative ideal theory

§8. The homomorphism theorem for rings

If $\mathfrak{a}$ is a two-sided ideal of $\mathfrak{o}$, then the residue class domain $\mathfrak{o}/\mathfrak{a}$ is not only an $\mathfrak{o}$-module but also a ring. Every homomorphic image of $\mathfrak{o}$ is again a ring and is isomorphic to a residue class ring $\mathfrak{o}/\mathfrak{a}$, where $\mathfrak{a}$ is a two-sided ideal. The proof is the same as for groups.

In particular, if $\mathfrak{o}$ is a division ring, then there are no ideals other than (0) and $\mathfrak{o}$; hence a homomorphism is either an isomorphism or sends every element to zero.

If the ring $\mathfrak{o}$ is a right K-module and K is a ring which commutes elementwise with $\mathfrak{o}$,

$$(ab)x = a(bx) = (ax)b,$$

for example if $\mathfrak{o}$ is a hypercomplex system relative to a commutative field K, then only those right, left, or two-sided ideals are admitted which are at the same time K-modules. Besides ring homomorphisms, one demands operator homomorphisms with respect to multiplication by K. The admissible ideals are bimodules relative to $\mathfrak{o}$ and K. In hypercomplex systems, the word "ideal" by itself always means such an ideal, admissible relative to the underlying coefficient field K. With this enlargement, the homomorphism theorem just stated remains valid.

The passage from a multiplier domain $\mathfrak{o}$ to the absolute multiplier domain (§1, Example 4) is an example of ring homomorphism. Thus the absolute multiplier domain is isomorphic to a residue class ring $\mathfrak{o}/\mathfrak{d}$, where $\mathfrak{d}$ is the two-sided ideal of all elements of $\mathfrak{o}$ which annihilate the module M.

From the homomorphism theorem one obtains, as in §2, the first and second isomorphism theorems for ring isomorphism and operator-isomorphism.

All products AB of subsets of $\mathfrak{o}$ are to be understood as module products: AB is the set of all sums of products ab with $a \in A$, $b \in B$; and aB is the set of all sums ab with $b \in B$. If B is a right module with respect to any ring as operator domain, then AB, respectively aB, is again a right module. In particular it may be a right ideal, an admissible ideal relative to a coefficient domain, and so on. If (A, B) denotes the sum of the modules A and B, the calculation rules are

$$A(B, C) = (AB, AC), \qquad (A, B)C = (AC, BC).$$

The direct sum is denoted by $A + B$.

In what follows we often consider rings satisfying a maximal or a minimal condition for right ideals. These include, in particular, the hypercomplex systems, since by convention only K-modules are admitted as ideals, and their ranks are therefore bounded.

§9. Idempotent elements. Direct-sum decomposition into right ideals

Let $\mathfrak{o}$ be a ring with unit element.

If $\mathfrak{r}$ is a right ideal, then $\mathfrak{r}\mathfrak{o} \subseteq \mathfrak{r}$; because there is a unit element, in fact $\mathfrak{r}\mathfrak{o} = \mathfrak{r}$. Similarly for left ideals, $\mathfrak{o}\mathfrak{l} = \mathfrak{l}$.

An element e is called **idempotent** if $e^2 = e$.

Let

$$\mathfrak{o} = \mathfrak{r}_1 + \cdots + \mathfrak{r}_s$$

be a decomposition into right ideals, and let

$$1 = e_1 + \cdots + e_s, \qquad e_i \in \mathfrak{r}_i.$$

Then

$$e_i e_k = 0 \quad (i \neq k), \qquad \mathfrak{r}_i = e_i \mathfrak{o}.$$

These are the orthogonality relations.

For if $r \in \mathfrak{r}_i$, then

$$r = re = re_1 + \cdots + re_s,$$

while also

$$r = 0 + \cdots + 0 + r + 0 + \cdots + 0.$$

Hence $re_k = 0$ for $k \neq i$, and $re_i = r$. The first formula gives $\mathfrak{r}_i \subseteq e_i\mathfrak{o}$, while the reverse inclusion is clear, so $\mathfrak{r}_i = e_i\mathfrak{o}$. Taking $r = e_i$ gives

$$e_i^2 = e_i, \qquad e_i e_k = 0 \quad (i \neq k).$$

Conversely, if

$$1 = e_1 + \cdots + e_s, \qquad e_i^2 = e_i, \qquad e_i e_k = 0 \ (i \neq k),$$

and if $\mathfrak{r}_i = e_i\mathfrak{o}$, then

$$\mathfrak{o} = \mathfrak{r}_1 + \cdots + \mathfrak{r}_s$$

directly. Every element r of $\mathfrak{o}$ is

$$r = e_1 r + \cdots + e_s r.$$

The representation is unique, since from

$$0 = e_1 a_1 + \cdots + e_s a_s$$

multiplication by e_i gives $e_i a_i = 0$.

From a right decomposition

$$\mathfrak{o} = e_1\mathfrak{o} + \cdots + e_s\mathfrak{o}$$

one obtains a left decomposition

$$\mathfrak{o} = \mathfrak{o}e_1 + \cdots + \mathfrak{o}e_s.$$

If the right ideals $\mathfrak{r}_i$ are directly indecomposable, then so are the corresponding left ideals $\mathfrak{l}_i = \mathfrak{o}e_i$; otherwise a decomposition of some $\mathfrak{l}_i$ would give a corresponding decomposition of $\mathfrak{r}_i$ by the same orthogonal idempotent calculation.

There is also the following converse form. If $e_i e_k = 0$ for $i \neq k$, if $e_i^2 = e_i$, and if $e = e_1 + \cdots + e_s$, then e is a left identity for the ring

$$e_1\mathfrak{o} + \cdots + e_s\mathfrak{o}.$$

The directness of the sum is proved as above.

§10. Decomposition into two-sided ideals

Assume again that $\mathfrak{o}$ is a ring with unit element, and that

$$\mathfrak{o} = \mathfrak{a}_1 + \cdots + \mathfrak{a}_s$$

is a decomposition of $\mathfrak{o}$ into two-sided ideals. Then the orthogonality relations hold also for the ideals:

$$\mathfrak{a}_i\mathfrak{a}_k = 0 \quad (i \neq k), \qquad \mathfrak{a}_i^2 = \mathfrak{a}_i.$$

Indeed, if $\mathfrak{b}_i$ is the sum of all $\mathfrak{a}_j$ with $j \neq i$, then $\mathfrak{o} = \mathfrak{a}_i + \mathfrak{b}_i$ and

$$\mathfrak{a}_i\mathfrak{b}_i \subseteq \mathfrak{a}_i \cap \mathfrak{b}_i = 0.$$

A fortiori $\mathfrak{a}_i\mathfrak{a}_k = 0$ for $i \neq k$, and

$$\mathfrak{a}_i\mathfrak{o} = \mathfrak{a}_i(\mathfrak{a}_i + \mathfrak{b}_i) = \mathfrak{a}_i^2.$$

Since $\mathfrak{a}_i\mathfrak{o} = \mathfrak{a}_i$, this gives $\mathfrak{a}_i^2 = \mathfrak{a}_i$.

Conversely, if a direct sum decomposition satisfies these orthogonality relations, then each summand $\mathfrak{a}_i$ is a two-sided ideal. Right ideals in the ring $\mathfrak{a}_i$ are right ideals in $\mathfrak{o}$.

If $\mathfrak{r}$ is a right ideal of $\mathfrak{o}$, then $\mathfrak{r}$ is the direct sum of the right ideals

$$\mathfrak{r}\mathfrak{a}_1, \dots, \mathfrak{r}\mathfrak{a}_s.$$

For

$$\mathfrak{r} = \mathfrak{r}\mathfrak{o} = (\mathfrak{r}\mathfrak{a}_1, \dots, \mathfrak{r}\mathfrak{a}_s),$$

and the orthogonality relations make the sum direct. If $\mathfrak{r}$ is directly indecomposable, it can have only one component, and therefore must be contained in one of the $\mathfrak{a}_i$.

These facts reduce the ideal theory of $\mathfrak{o}$ to the ideal theory of the individual $\mathfrak{a}_i$. For this reason we shall usually restrict ourselves to rings which are two-sided indecomposable.

Two right ideals contained in different $\mathfrak{a}_i$ can never be operator-isomorphic. An ideal contained in $\mathfrak{a}_i$ is annihilated by all the other $\mathfrak{a}_j$, but not by $\mathfrak{a}_i$; the opposite holds for an ideal contained in $\mathfrak{a}_j$.

If a ring can be decomposed as

$$\mathfrak{o} = \mathfrak{a}_1 + \cdots + \mathfrak{a}_s$$

with the $\mathfrak{a}_i$ two-sided indecomposable, and if each $\mathfrak{a}_i$ is decomposed into one-sided indecomposable right ideals, then operator-isomorphic right ideals are to be looked for only among those belonging to the same $\mathfrak{a}_i$. It may nevertheless happen that there are several classes of non-isomorphic right ideals within the same $\mathfrak{a}_i$.

For example, let K be the field of rational numbers and let

$$\mathfrak{o} = e_1K + e_2K + uK$$

be a hypercomplex system in which e_1, e_2, u commute with the elements of K, and multiplication is given by

$$\begin{array}{c|ccc} & e_1 & e_2 & u \\ \hline e_1 & e_1 & 0 & u \\ e_2 & 0 & e_2 & 0 \\ u & u & 0 & 0 \end{array}$$

The unit element is $e = e_1 + e_2$. The e_i satisfy the orthogonality relations. Thus the right decomposition is

$$\mathfrak{o} = e_1\mathfrak{o} + e_2\mathfrak{o} = (e_1, u) + (e_2),$$

and the left decomposition is

$$\mathfrak{o} = \mathfrak{o}e_1 + \mathfrak{o}e_2 = (e_1) + (e_2, u).$$

Since $e_1\mathfrak{o}$ and $\mathfrak{o}e_1$ are visibly indecomposable, the corresponding opposite one-sided ideals are indecomposable as well. A two-sided decomposition is impossible, since one component would have to contain $e_1\mathfrak{o}$ and the other $e_2\mathfrak{o}$; then the first would contain u, but the second would contain $ue_1 = u$ as well. Yet $e_1\mathfrak{o}$ and $e_2\mathfrak{o}$ are not operator-isomorphic, since they have different ranks over K.

Now suppose

$$\mathfrak{o} = \mathfrak{a}_1 + \cdots + \mathfrak{a}_s$$

with the $\mathfrak{a}_i$ two-sided indecomposable. Then the $\mathfrak{a}_i$ are uniquely determined. Indeed, suppose also

$$\mathfrak{o} = \mathfrak{c}_1 + \cdots + \mathfrak{c}_t.$$

Then

$$\mathfrak{a}_i = \mathfrak{a}_i\mathfrak{o} = (\mathfrak{a}_i\mathfrak{c}_1, \dots, \mathfrak{a}_i\mathfrak{c}_t),$$

and this sum is direct, since $\mathfrak{a}_i\mathfrak{c}_j \subseteq \mathfrak{c}_j$. Since $\mathfrak{a}_i$ is indecomposable, all these summands vanish except one. Thus $\mathfrak{a}_i = \mathfrak{a}_i\mathfrak{c}_j$. Symmetrically each $\mathfrak{c}_j$ contains precisely one of the $\mathfrak{a}_i$, and the two decompositions coincide up to order.

§11. The center

The **center** Z of a ring $\mathfrak{o}$ is the set of all elements z commuting with every ring element:

$$za = az \qquad (a \in \mathfrak{o}).$$

It is a commutative ring.

Every one-sided ideal $\mathfrak{a}$ of $\mathfrak{o}$ has a contracted ideal $\mathfrak{a} \cap Z$ in Z, since both $\mathfrak{a}$ and Z are Z-modules.

Every ideal A in Z has an extended ideal: the ideal generated by A in $\mathfrak{o}$,

$$\mathfrak{a} = (A, \mathfrak{o}A).$$

This ideal is two-sided, since

$$\mathfrak{a}\mathfrak{o} = (\mathfrak{o}A, A\mathfrak{o})\mathfrak{o} = (\mathfrak{o}A\mathfrak{o}, A\mathfrak{o}^2) \subseteq (\mathfrak{o}A, A\mathfrak{o}) = \mathfrak{a}.$$

When $\mathfrak{o}$ has a unit element one simply writes $\mathfrak{a} = \mathfrak{o}A$.

In that case the following theorem holds. Every two-sided decomposition of $\mathfrak{o}$ corresponds one-to-one to a decomposition of the center. From

$$\mathfrak{o} = \mathfrak{a}_1 + \cdots + \mathfrak{a}_s, \qquad A_i = \mathfrak{a}_i \cap Z$$

one obtains

$$Z = A_1 + \cdots + A_s, \qquad \mathfrak{a}_i = \mathfrak{o}A_i,$$

and conversely.

For let $z \in Z$ and write its decomposition with respect to the two-sided decomposition of $\mathfrak{o}$:

$$z = z_1 + \cdots + z_s.$$

Then the z_i are again central. Indeed, for all a,

$$za = z_1 a + \cdots + z_s a = az = az_1 + \cdots + az_s,$$

and comparison of components gives $z_i a = a z_i$, since the $\mathfrak{a}_i$ are two-sided. Hence $z_i \in \mathfrak{a}_i \cap Z = A_i$, and the above is the asserted direct-sum decomposition of Z. In particular, if

$$1 = e_1 + \cdots + e_s$$

is the decomposition of the unit, then the e_i are central elements. Finally

$$\mathfrak{o}A_i = \mathfrak{o}Ze_i = \mathfrak{o}e_i = \mathfrak{a}_i.$$

Conversely, let

$$Z = A_1 + \cdots + A_s$$

be a decomposition of the center. By §9 the components of the identity satisfy

$$1 = e_1 + \cdots + e_s, \qquad e_i^2 = e_i, \qquad e_i e_j = 0 \ (i \neq j).$$

The orthogonality relations give

$$\mathfrak{o} = \mathfrak{o}e_1 + \cdots + \mathfrak{o}e_s.$$

Putting $\mathfrak{a}_i = \mathfrak{o}A_i$ gives, by the first part of the proof,

$$A_i = \mathfrak{a}_i \cap Z.$$

It follows that $\mathfrak{a}_i = \mathfrak{o}e_i$, and the decomposition is the desired two-sided decomposition.

Consequently $\mathfrak{o}$ and Z are at the same time two-sided directly decomposable, or not.

§12. Nilpotent ideals

An ideal $\mathfrak{c}$ is called **nilpotent** if $\mathfrak{c}^r = 0$ for some r.

The ideal (u) in the example of §10 is nilpotent.

If $\mathfrak{o}$ contains a nilpotent right ideal $\mathfrak{r}$, then it contains a nilpotent left ideal, indeed a two-sided nilpotent ideal. If $\mathfrak{r}^r = 0$, then $\mathfrak{o}\mathfrak{r}$ is a left ideal; if $\mathfrak{o}\mathfrak{r}$ could be zero because $\mathfrak{o}$ has no unit element, take instead $(\mathfrak{o}\mathfrak{r}, \mathfrak{r})$. This ideal is nilpotent.

The sum of two nilpotent right ideals is again nilpotent. If $\mathfrak{c}^r = 0$ and $\mathfrak{d}^s = 0$, then in

$$(\mathfrak{c}, \mathfrak{d})^{r+s-1}$$

every term contains either at least r factors from $\mathfrak{c}$ or at least s factors from $\mathfrak{d}$, and so vanishes. Hence the whole power is zero.

If the maximal condition for right ideals holds in $\mathfrak{o}$, there is therefore a maximal nilpotent right ideal. It contains all other nilpotent right ideals, for otherwise one could form the sum with such an ideal. Call it $\mathfrak{c}$. Since $\mathfrak{oc}$ is again a nilpotent right ideal, $\mathfrak{oc} \subseteq \mathfrak{c}$, and therefore $\mathfrak{c}$ is a left ideal as well. Every nilpotent left ideal $\mathfrak{d}$ is contained in $\mathfrak{c}$, because $(\mathfrak{d}, \mathfrak{do})$ is a nilpotent right ideal. Thus there is a maximal two-sided nilpotent ideal $\mathfrak{c}$, the **radical**, which contains all nilpotent right and left ideals.

In the example of §10, (u) is the maximal nilpotent ideal.

If the radical is the zero ideal, one speaks of a **ring without radical**: equivalently, a semisimple ring, or a Dedekind system. The residue class ring modulo the radical is always a ring without radical.

Let Z be the center of $\mathfrak{o}$ and $\mathfrak{c}$ the radical of $\mathfrak{o}$. Then

$$Z \cap \mathfrak{c}$$

is the radical of Z. It is plainly nilpotent. If there were a larger nilpotent ideal C in Z, it would extend to a nilpotent ideal of $\mathfrak{o}$ not wholly contained in $\mathfrak{c}$:

$$(C\mathfrak{o})^r = C^r \mathfrak{o}^r = 0.$$

This is impossible by maximality of $\mathfrak{c}$.

It follows that if $\mathfrak{o}$ is without radical, then so is its center. The converse is false, as the example in §10 shows. The center of $\mathfrak{o}/\mathfrak{c}$ contains a ring isomorphic to $Z/(Z \cap \mathfrak{c})$, but this ring may be a proper subring.

§13. Completely reducible rings

By §5, a ring is called **right completely reducible** if it is the direct sum of finitely many simple right ideals.

A right completely reducible ring with unit element has no non-zero nilpotent ideal, hence no radical. Indeed every right ideal $\mathfrak{r}$ is a direct summand:

$$\mathfrak{o} = \mathfrak{r} + \mathfrak{s} = e_1\mathfrak{o} + e_2\mathfrak{o}.$$

Since $e_1^2 = e_1$, one also has $e_1 = e_1^r$. If $\mathfrak{r}$ were nilpotent, $e_1^r = 0$ and so $e_1 = 0$, whence $\mathfrak{r} = 0$.

We now prove the two converses.

First: A ring without radical satisfying the minimal condition for right ideals is completely reducible relative to right ideals.

Let $\mathfrak{t} \neq 0$ be a minimal right ideal. We first show that $\mathfrak{t}$ contains an idempotent. Since $\mathfrak{t}^2 \subseteq \mathfrak{t}$ and $\mathfrak{t}^2 \neq 0$, we have $\mathfrak{t}^2 = \mathfrak{t}$. Thus there is an $a \in \mathfrak{t}$ with $a\mathfrak{t} \neq 0$, and therefore $a\mathfrak{t} = \mathfrak{t}$. The set of all $b \in \mathfrak{t}$ annihilated by a, so that $ab = 0$, is a right ideal. It is not all of $\mathfrak{t}$, hence it is zero. Therefore $ab = 0$ implies $b = 0$.

Because $\mathfrak{t} = a\mathfrak{t}$, one may write $a = ac$ with $c \neq 0$. It follows that

$$a(c^2 - c) = 0,$$

and therefore $c^2 = c$.

We next show that $\mathfrak{t}$ is a direct summand. The right ideal $c\mathfrak{o}$ is contained in $\mathfrak{t}$ and is non-zero because $c^2 = c$ lies in it; hence $c\mathfrak{o} = \mathfrak{t}$. Every element $r \in \mathfrak{o}$ has the Peirce decomposition

$$r = cr + (r - cr).$$

The elements cr form $\mathfrak{t}$. The elements $r - cr$ form another right ideal, annihilated by c:

$$c(r - cr) = 0.$$

Thus

$$\mathfrak{o} = \mathfrak{t} + \mathfrak{r}.$$

Because the elements of $\mathfrak{t}$ are not annihilated by c, the sum is direct.

Now decompose $\mathfrak{o}$, using the minimal condition, as a direct sum of directly indecomposable ideals,

$$\mathfrak{o} = \mathfrak{r}_1 + \cdots + \mathfrak{r}_s.$$

The $\mathfrak{r}_i$ must be simple, equivalently minimal. If, for instance, $\mathfrak{r}_1$ were not simple and $\mathfrak{t}$ were a minimal ideal in $\mathfrak{r}_1$, the direct-summand result above would give a decomposition of $\mathfrak{r}_1$, contradiction. Thus $\mathfrak{o}$ is completely reducible.

Second: A ring without radical satisfying the minimal condition has a unit element.

To establish first the existence of a left unit, it suffices to prove the following. If a non-zero right ideal $\mathfrak{a} = e_1\mathfrak{o}$ has a left unit e_1, then there is a right ideal $\mathfrak{b}$ of larger length which also has a left unit e. Since complete reducibility, hence boundedness of all lengths, has already been proved, finitely many repetitions yield a left unit for $\mathfrak{o}$ itself.

Assume $\mathfrak{a} = e_1\mathfrak{o}$ and $e_1^2 = e_1$. The formula

$$r = e_1 r + (r - e_1 r)$$

gives a Peirce decomposition $\mathfrak{o} = e_1\mathfrak{o} + \mathfrak{r}$. Let e_2 be an idempotent in $\mathfrak{r}$ as above. Then $e_1 e_2 = 0$, because e_1 annihilates all elements of $\mathfrak{r}$. Put

$$e_2' = e_2 - e_2 e_1.$$

Then

$$(e_2')^2 = e_2', \qquad e_2' e_1 = 0, \qquad e_1 e_2' = 0,$$

so $e_2'\mathfrak{o}$ is non-zero; and $e = e_1 + e_2'$ is a left unit for the ring $e_1\mathfrak{o} + e_2'\mathfrak{o}$, which has larger right-ideal length than $\mathfrak{a}$.

To show that the left unit thus constructed is also a right unit, make the Peirce decomposition into left ideals:

$$\mathfrak{o} = \mathfrak{o}e + \mathfrak{l}.$$

We have $\mathfrak{l}e = 0$ and $\mathfrak{l} = e\mathfrak{l}$, so

$$\mathfrak{l}^2 = \mathfrak{l}e\mathfrak{l} = 0.$$

Since by hypothesis no nilpotent left ideal can exist, $\mathfrak{l} = 0$. Thus e is a right unit as well, and hence a unit element.

Theorem. From right complete reducibility and the existence of a unit element follows two-sided complete reducibility:

$$\mathfrak{o} = \mathfrak{d}_1 + \cdots + \mathfrak{d}_s,$$

where the $\mathfrak{d}_i$ are two-sided simple.

As before, it is enough to show that every minimal two-sided ideal $\mathfrak{a}$ is a direct summand. As a right ideal,

$$\mathfrak{o} = \mathfrak{a} + \mathfrak{r} = e_1\mathfrak{o} + e_2\mathfrak{o}.$$

The corresponding left decomposition is

$$\mathfrak{o} = \mathfrak{o}e_1 + \mathfrak{o}e_2 = \mathfrak{a} + \mathfrak{l}.$$

One obtains

$$\mathfrak{a}\mathfrak{l} = 0, \qquad (\mathfrak{a}\mathfrak{l})^2 = 0,$$

and hence $\mathfrak{a}\mathfrak{l} = 0$; similarly $\mathfrak{l}\mathfrak{a} = 0$. It follows that $\mathfrak{r} = \mathfrak{l}$ and is two-sided, so $\mathfrak{a}$ is a two-sided direct summand. The rest of the proof is the same as for one-sided complete reducibility.

The $\mathfrak{d}_i$ are two-sided simple also as rings, because all ideals in $\mathfrak{d}_i$ are ideals in $\mathfrak{o}$. We now examine their structure.

§14. Two-sided simple completely reducible rings with unit element

Let $\mathfrak{o}$ be such a ring, and let

$$\mathfrak{o} = \mathfrak{r}_1 + \cdots + \mathfrak{r}_n = e_1\mathfrak{o} + \cdots + e_n\mathfrak{o}$$

be a decomposition into simple right ideals. Then

$$\mathfrak{o} = \mathfrak{l}_1 + \cdots + \mathfrak{l}_n = \mathfrak{o}e_1 + \cdots + \mathfrak{o}e_n,$$

where the $\mathfrak{l}_i$ are directly indecomposable left ideals.

The product $\mathfrak{l}_i\mathfrak{r}_j$ is a non-zero two-sided ideal whenever i, j are fixed and the product is non-zero; since $\mathfrak{o}$ is two-sided simple, one has, in fact, enough non-zero components to recover all of $\mathfrak{o}$. Put

$$A_{ij} = \mathfrak{l}_i\mathfrak{r}_j.$$

Then

$$\mathfrak{o} = \sum_{i,j} A_{ij}$$

directly: from

$$\sum a_{ij} = 0, \qquad a_{ij} \in A_{ij},$$

one first uses the directness of the right decomposition and then the directness of the left decomposition to conclude each $a_{ij} = 0$.

Moreover

$$A_{ij}\mathfrak{r}_j = A_{ij} \neq 0,$$

and for any non-zero $a_{ij} \in A_{ij}$ one has

$$a_{ij}\mathfrak{r}_j \subseteq \mathfrak{r}_i, \qquad a_{ij}\mathfrak{r}_j \neq 0.$$

Since $\mathfrak{r}_j$ is simple, multiplication by a_{ij} gives an operator homomorphism

$$\mathfrak{r}_j \to \mathfrak{r}_i.$$

The kernel is an ideal in $\mathfrak{r}_j$, hence zero. Thus it is an isomorphism. Therefore any two simple right ideals occurring in a decomposition of $\mathfrak{o}$ are operator-isomorphic; the isomorphisms are mediated by elements of A_{ij}. Since any two simple right ideals can be made to occur together in some decomposition of $\mathfrak{o}$, all simple right ideals are mutually isomorphic.

All homomorphisms from $\mathfrak{r}_j$ to $\mathfrak{r}_i$ are mediated by elements of A_{ij}. In particular, endomorphisms of $\mathfrak{r}_i$ are mediated by elements of A_{ii}. The product and sum of elements of A_{ii} correspond to product and sum of endomorphisms. Distinct elements give distinct endomorphisms, since e_i goes to $a_{ii}e_i = a_{ii}$. Hence the ring A_{ii} is isomorphic to the endomorphism ring of $\mathfrak{r}_i$.

But the endomorphism ring of a simple ideal is a division ring. Therefore A_{ii} is a division ring. Since all $\mathfrak{r}_i$ are operator-isomorphic, their endomorphism rings are ring-isomorphic. Thus A_{ii} is determined by $\mathfrak{o}$ uniquely up to ring isomorphism. The various possible A_{ii} are merely different concrete realizations of the abstract endomorphism division ring of the simple right ideals.

Main theorem. Every completely reducible two-sided simple ring $\mathfrak{o}$ with unit element is isomorphic to the ring of all n by n matrices over a division ring K. The division ring K is isomorphic to the automorphism division ring of the right ideals of $\mathfrak{o}$.

We construct matrix units. Let I_i be the identity automorphism of $\mathfrak{r}_i$, and let I_{ij} be an arbitrary isomorphism $\mathfrak{r}_j \to \mathfrak{r}_i$. Put

$$I_{ik} = I_{ij} I_{jk}.$$

If I_{ij} is mediated by an element $c_{ij} \in A_{ij}$, then

$$c_{ij} = e_i c_{ij} = c_{ij} e_j,$$

and

$$c_{ij} c_{jk} = c_{ik}, \qquad c_{ij} c_{hk} = 0 \quad (j \neq h).$$

The c_{ij} are therefore matrix units.

Now define K. To an element $a_{11} \in A_{11}$ assign its conjugates

$$a_{ii} = c_{i1} a_{11} c_{1i} \in A_{ii},$$

and set

$$\alpha = a_{11} + a_{22} + \cdots + a_{nn}.$$

The correspondence $a_{11} \mapsto \alpha$ is a ring isomorphism from A_{11} onto the set of all such α; call this set K. Every $\alpha \in K$ commutes with all matrix units c_{ij}. Moreover

$$\mathfrak{o} = \sum_{i,j} c_{ij} K.$$

Thus every element of $\mathfrak{o}$ can be written uniquely in the form

$$\sum_{i,j} c_{ij} \alpha_{ij},$$

and assigning to it the matrix (α_{ij}) gives an isomorphism of $\mathfrak{o}$ onto the full matrix ring over K. This proves the theorem.

Conversely, the ring of all n by n matrices over a division ring A is two-sided simple and completely reducible; A is isomorphic to the automorphism ring of the right ideals, and the construction above recovers the subring $K = A$.

Let c_{ij} be the usual matrix units, and put $\mathfrak{r}_i = c_{ii}\mathfrak{o}$. Then

$$\mathfrak{o} = \mathfrak{r}_1 + \cdots + \mathfrak{r}_n.$$

The $\mathfrak{r}_i$ are simple right ideals, since every non-zero element $\sum_j c_{ij}\alpha_j$ generates the whole $\mathfrak{r}_i$. Indeed, if $\alpha_k \neq 0$, then multiplication by $\alpha_k^{-1}c_{ki}$ produces c_{ii}, and $c_{ii}\beta$ runs through all elements of $\mathfrak{r}_i$.

The right ideals $\mathfrak{r}_i$ are operator-isomorphic by $\mathfrak{r}_i \mapsto c_{ji}\mathfrak{r}_i$. It follows that the ring is two-sided indecomposable; by the theorem of §13, since it is already right completely reducible and has a unit, it is two-sided simple. Equivalently, every non-zero element generates the full two-sided ideal.

The center of $\mathfrak{o}$ is the center of K, and hence is a field. Indeed, a central element in the matrix ring must commute with all matrix units, so it has the form

$$\alpha c_{11} + \cdots + \alpha c_{nn},$$

where α lies in the center of K. Conversely, every such scalar diagonal matrix is central.

Since a matrix ring is also left completely reducible, and every right completely reducible ring with unit is a direct sum of matrix rings, it follows that every right completely reducible ring with unit is also left completely reducible, and conversely. The center of a completely reducible ring with unit is a direct sum of commutative fields corresponding to the two-sided simple matrix-ring components.

We shall also use this fact: any two decompositions

$$\mathfrak{o} = \sum c_{ij}K, \qquad \mathfrak{o} = \sum c'_{ij}K'$$

are carried into each other by an inner automorphism,

$$x \mapsto y^{-1}xy, \qquad c'_{ij} = y^{-1}c_{ij}y.$$

The proof is by choosing elements mediating reciprocal isomorphisms between corresponding right ideals and forming the usual conjugating element.

Chapter III. Module and representation theory

§15. Representations and representation modules

Let $\mathfrak{o}$ be a ring and K a ring with unit element. In the later applications K will always be a field.

A representation of degree n of $\mathfrak{o}$ in K is a homomorphism

$$\mathfrak{o} \to D,$$

where D is a ring of n by n matrices with entries in K.

A **representation module** of $\mathfrak{o}$ relative to K is a bimodule M which is a left $\mathfrak{o}$-module and a right K-module:

$$\mathfrak{o}M \subseteq M, \qquad MK \subseteq M,$$

which is a direct sum of finitely many one-generator K-modules,

$$M = x_1 K + \cdots + x_n K,$$

and in which the unit element of K is the identity operator.

Every representation module gives a representation. If $c \in \mathfrak{o}$ and

$$c x_i = \sum_j x_j \gamma_{ji},$$

then, in matrix notation,

$$c(x_1, x_n) = (x_1, \ldots, x_n)C, \qquad C = (\gamma_{ji}).$$

The matrices C form a representation, for

$$(b+c)x_i = bx_i + cx_i, \qquad (bc)x_i = b(cx_i),$$

and these are exactly the matrix rules $B + C$ and BC.

Conversely, every representation D of $\mathfrak{o}$ belongs to a representation module, and indeed to a particular basis of that module. Let M be the set of formal linear forms

$$y = x_1\alpha_1 + \cdots + x_n\alpha_n, \qquad \alpha_i \in K.$$

Then M is a right K-module. If the element c is represented by the matrix $C = (\gamma_{ji})$, define

$$c x_i = \sum_j x_j \gamma_{ji},$$

and extend by linearity. The homomorphism relations for the matrices give

$$(c+d)y = cy + dy, \qquad (cd)y = c(dy),$$

and the remaining bimodule laws

$$c(y+z) = cy + cz, \qquad c(y\alpha) = (cy)\alpha$$

are immediate. Thus M is a representation module, and, relative to the chosen basis, it gives exactly the representation $\mathfrak{o} \to D$.

If $(y_1, \ldots, y_n)$ is another basis of the same representation module and

$$(y_1, \ldots, y_n) = (x_1, \ldots, x_n)P,$$

then the matrices obtained from this new basis are

$$P^{-1}CP.$$

Representations related by

$$C \mapsto P^{-1}CP$$

are called equivalent and are counted in the same representation class. Since every regular matrix P transforms one basis of M into another, every representation module determines uniquely a representation class.

Equivalently, one may formulate the correspondence invariantly, without a special choice of basis. A representation module assigns to the multiplier domain $\mathfrak{o}$ a homomorphic system of linear transformations of the module. Choosing a basis only writes these transformations as matrices. Conversely, any homomorphic system of linear transformations of a finite linear-form module gives a representation module. Thus the relation between modules and representation classes is independent of coordinates.

It is clear that operator-isomorphic representation modules determine the same representation class. Conversely, if two representation modules with bases (x_i) and (z_i) produce the same representation, then

$$x_i \mapsto z_i$$

extends to an operator-isomorphism. The multiplication law is completely determined by the matrices C.

In particular, if two bases of the same module produce the same representation, then they differ by an operator-isomorphism of the module onto itself. Equivalently, if

$$C = P^{-1}CP$$

for every C and one fixed P, then the transformation of the module induced by P is an automorphism. Conversely, every automorphism of the bimodule is mediated by a matrix commuting with all matrices of the representation.

The notion of representation can also be extended to rings which are commutatively connected with a commutative multiplier domain P. In that case the representing rings K must contain P in their center, and a representation is required to be not only a ring homomorphism but also an operator homomorphism: from $a \mapsto A$ one must have

$$a\rho \mapsto A\rho \qquad (\rho \in P).$$

For representation modules this gives the additional rule

$$am \cdot \rho = a(m\rho) = a\rho \cdot m.$$

One says that the module and the ring are commutatively connected with P.

§16. Reducible representations

Let A be a submodule of the representation module M, and suppose it is possible to choose a basis for M consisting of a basis $z_1, \dots, z_t$ of A, completed by elements $y_1, \dots, y_r$:

$$M = y_1K + \cdots + y_rK + z_1K + \cdots + z_tK.$$

Then the representing matrices have block triangular form

$$C = \begin{pmatrix} R & S \\ 0 & T \end{pmatrix}.$$

The matrices T alone form a representation of degree t, mediated by A, and the matrices R form a representation of degree r, mediated by M/A.

For the products cz_i are again elements of A, so they are expressible in terms of the z_i alone; hence the lower-left block is zero in the chosen order. The remaining blocks give the two induced representations. Conversely, if a representation is given in such a block form, then the subspace generated by the last t basis elements is a submodule.

Let

$$M = A_0 \supset A_1 \supset \cdots \supset A_s = 0$$

be a composition series for M, and suppose that each A_i is a direct summand of the preceding module as a K-module. Then one can choose a basis adapted to the series, and the matrices have the form

$$\begin{pmatrix} R_{11} & * & * & \cdots & * \\ 0 & R_{22} & * & \cdots & * \\ 0 & 0 & R_{33} & \cdots & * \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & R_{ss} \end{pmatrix}.$$

The diagonal blocks R_{ii} are representations mediated by the composition factors

$$A_{i-1}/A_i.$$

These representations are irreducible, since the composition factors are simple.

If K is a field, every submodule is a direct summand, and conversely every representation reduced as far as possible in the above sense gives a composition series. By the Jordan-Hölder theorem, the composition factors are uniquely determined up to operator-isomorphism; hence the diagonal representations R_{ii} are uniquely determined up to equivalence and order.

If

$$M = A + B$$

is a direct sum of representation modules, then with a basis adapted to the two summands the representation has block diagonal form

$$\begin{pmatrix} R & 0 \\ 0 & S \end{pmatrix},$$

where R is mediated by A and S by B. Conversely, such a block diagonal form corresponds to a direct-sum decomposition of the representation module.

If one first decomposes M as a direct sum of directly indecomposable constituents and then draws composition series in these constituents, the representation can be put into a block form in which the large diagonal boxes correspond to the directly indecomposable summands, and inside each box the diagonal pieces correspond to the composition factors. If K is a field, every representation reduced as far as possible in this way conversely gives such a module decomposition. By the theorem of Krull and Otto Schmidt, the classes of the directly indecomposable constituents are uniquely determined up to order.

If the module M is completely reducible, all boxes consist of single irreducible representations. The corresponding representation is called **completely reducible**.

§17. Direct-sum decomposition of representation modules for rings with unit element

Let M be an $\mathfrak{o}$-module, with $\mathfrak{o}$ a ring with unit element. Then

$$M = M_e + M_0,$$

where the unit is the identity operator on M_e and the zero operator on M_0. If M is also a right K-module, then M_e and M_0 are right K-modules as well.

Every $m \in M$ is decomposed as

$$m = em + (m - em).$$

Let M_e be the set of all em, and M_0 the set of all $m - em$. These are additive groups, and right K-modules if M is one. Moreover

$$r(em) = erm,$$

so M_e is an $\mathfrak{o}$-module; also

$$r(m - em) = rm - erm = rm - rm = 0,$$

so M_0 is an $\mathfrak{o}$-module. The unit e acts as the identity on the elements em and as zero on the elements $m - em$. Hence only the zero element belongs to both M_e and M_0, and the sum is direct.

For representation modules M_0 gives the trivial representation, in which every element is assigned the zero matrix. After splitting off M_0 there remains the representation mediated by M_e, in which the unit element is represented by the identity matrix. From now on we may and shall restrict ourselves to such representations.

Let

$$\mathfrak{o} = \mathfrak{a}_1 + \cdots + \mathfrak{a}_s$$

be a direct sum of two-sided ideals, let

$$e = e_1 + \cdots + e_s$$

be the corresponding decomposition of the unit, and let M be an $\mathfrak{o}$-module on which e is the identity operator. Then

$$M = \mathfrak{a}_1 M + \cdots + \mathfrak{a}_s M$$

directly.

Indeed $M = \mathfrak{o}M = (\mathfrak{a}_1 M, \ldots, \mathfrak{a}_s M)$. If $\mathfrak{b}_i$ denotes the sum of all components except $\mathfrak{a}_i$, then

$$M = (\mathfrak{a}_i M, \mathfrak{b}_i M),$$

and this sum is direct because e_i acts as identity on $\mathfrak{a}_i M$ and as zero on $\mathfrak{b}_i M$.

We shall therefore usually restrict ourselves to representations of two-sided indecomposable rings.

§18. Module and representation theory of completely reducible rings

Let $\mathfrak{o}$ be completely reducible and two-sided simple, and let M be a finite $\mathfrak{o}$-module on which the unit of $\mathfrak{o}$ is the identity operator. Then M is completely reducible, and its simple constituents are operator-isomorphic to the simple left ideals of $\mathfrak{o}$.

For let

$$\mathfrak{o} = \mathfrak{l}_1 + \cdots + \mathfrak{l}_n$$

be the decomposition into simple left ideals, and suppose

$$M = (\mathfrak{o}m_1, \ldots, \mathfrak{o}m_r).$$

Then

$$\mathfrak{o}m_j = (\mathfrak{l}_1 m_j, \ldots, \mathfrak{l}_n m_j).$$

The non-zero modules $\mathfrak{l}_i m_j$ are isomorphic to $\mathfrak{l}_i$, since the map $a \mapsto am_j$ is an operator homomorphism whose kernel is a left ideal. Omitting those summands already contained in previous ones, one obtains a direct sum of simple modules. Thus M is completely reducible. If M is directly indecomposable, it is simple and isomorphic to one of the $\mathfrak{l}_i$.

Let A be the automorphism division ring of such a simple module M, and let A_0 be that of $\mathfrak{l}_1$. Since M and $\mathfrak{l}_1$ are operator-isomorphic, the two division rings are ring-isomorphic. By §1, both M and $\mathfrak{l}_1$ may be regarded as bimodules with these automorphism division rings on the right. They are representation modules: $\mathfrak{l}_1$, and therefore M, has finite rank over its automorphism division ring. The representations mediated by them correspond under the ring isomorphism $A \simeq A_0$. Thus the study of this representation class may be reduced to the representation mediated by $\mathfrak{l}_1$ in its own automorphism division ring.

Choose for $\mathfrak{l}_1$ a basis consisting of matrix units

$$\mathfrak{l}_1 = (c_{11}, \ldots, c_{n1})K,$$

where K denotes the concrete realization, from §14, of the automorphism division ring. Write every element $a \in \mathfrak{o}$ as

$$a = \sum_{i,j} c_{ij}\alpha_{ij}.$$

Then the representation mediated by $\mathfrak{l}_1$ over K is

$$a \mapsto (\alpha_{ij}).$$

Indeed, multiplication of a on the basis column $(c_{11}, \ldots, c_{n1})$ gives the corresponding matrix of coefficients.

Now let Γ be a subfield of K such that K has finite rank over Γ:

$$K = x_1\Gamma + \cdots + x_t\Gamma.$$

Then K itself is a representation module over Γ. Each element $\beta \in K$ is represented over Γ by a matrix $B = (\beta_{ij})$ determined by

$$\beta x_i = \sum_j x_j \beta_{ji}.$$

Regarding $\mathfrak{l}_1$ as representation module relative to Γ, the representation of $\mathfrak{o}$ over Γ is obtained from

$$a \mapsto (\alpha_{ij})$$

by replacing each entry α_{ij} by the matrix A_{ij} which represents that element of K over Γ.

The representations just studied - those in the automorphism division ring of the left ideals and in its finite subfields - are, up to isomorphism, the only ones that can be mediated by simple $\mathfrak{o}$-modules, or by left ideals. For if one regards an $\mathfrak{o}$-module M as a bimodule relative to some field K, then the elements of K induce module homomorphisms. Hence K must be mapped isomorphically into a subfield of the automorphism division ring of M. The full automorphism division ring must have finite degree over that subfield, otherwise the representation module would have infinite rank.

These irreducible representations are not yet all irreducible representations in general. There may be representation modules which are reducible as $\mathfrak{o}$-modules but irreducible as bimodules, and which therefore still yield irreducible representations.

§19. The simple composition factors in modules and representation modules

Let $\mathfrak{o}$ be a ring satisfying the maximal and minimal conditions for left ideals, and let $\mathfrak{c}$ be its maximal nilpotent ideal. Then every simple $\mathfrak{o}$-module M is either annihilated by $\mathfrak{o}$, or else it is isomorphic to a simple left ideal $\mathfrak{l}$ of the radical-free ring $\mathfrak{o}/\mathfrak{c}$. In the latter case the module is annihilated by precisely those two-sided ideals of $\mathfrak{o}/\mathfrak{c}$ which do not contain $\mathfrak{l}$, and its absolute multiplier domain is isomorphic to the two-sided simple ring containing $\mathfrak{l}$.

If $\mathfrak{c}^r = 0$, then necessarily $\mathfrak{c}M = 0$; otherwise

$$M = \mathfrak{c}M = \mathfrak{c}^2 M = \cdots = 0,$$

a contradiction. Thus M may be regarded as an $\mathfrak{o}/\mathfrak{c}$-module. The ring $\mathfrak{o}/\mathfrak{c}$ has a unit element and no radical. Hence M decomposes into one part annihilated by $\mathfrak{o}/\mathfrak{c}$ and one part on which the unit is the identity; since M is simple, only one part can occur. If the unit is the identity, the argument of §18 gives an isomorphism to a simple left ideal. The remaining assertions are clear for such left ideals and pass immediately to all modules isomorphic to them.

Consequently, if M is an $\mathfrak{o}$-module with a composition series, its simple composition factors are either annihilated by $\mathfrak{o}$ or are isomorphic to simple left ideals of $\mathfrak{o}/\mathfrak{c}$.

If P is a ring commutatively connected with $\mathfrak{o}$, all results remain valid after replacing modules by bimodules relative to P on the right and $\mathfrak{o}$ on the left, subject to

$$am \cdot \rho = a(m\rho) = a\rho \cdot m.$$

The unit element of P is to be the identity operator. Submodules are always required to admit multiplication by P. The submodules constructed in the proofs, such as $\mathfrak{c}M, \mathfrak{c}^2M, \ldots$, have this property.

In particular, if P is a field and M is a finite-rank P-module, then M is a representation module. The representation mediated by M is a ring homomorphism and also an operator homomorphism with respect to P. Conversely, all such representations over P arise from such modules. Irreducible representations correspond to simple modules, and conversely. Therefore the preceding theorems may be read as theorems about representations of $\mathfrak{o}$ over P: apart from the zero representations, all irreducible representations are mediated by the simple left ideals of $\mathfrak{o}/\mathfrak{c}$. If an arbitrary representation is reduced by means of a composition series as in §16, then the only diagonal representations which occur, besides zeros, are equivalent to those mediated by simple left ideals of $\mathfrak{o}/\mathfrak{c}$.

For these statements, no finiteness condition on $\mathfrak{o}$ is needed beyond the hypotheses already placed on the representation over P. Every representation is an isomorphic representation of its absolute multiplier domain. This absolute multiplier domain is, as the inverse image of a finite matrix ring, itself a finite-rank P-module. Since the admissible ideals are by definition P-modules, the maximal and minimal conditions hold for it.

If P is a commutative field, this absolute multiplier domain is a hypercomplex system over P. This assumption is automatically fulfilled whenever non-zero representations occur on the diagonal: in that case P is isomorphic to a subfield of the automorphism division ring of a simple left ideal, and, in the realization by the subfield K of $\mathfrak{o}/\mathfrak{c}$, the commutative connection forces it to lie in the center of K.

Chapter IV. Representations of groups and hypercomplex systems

§20. Inclusion of hypercomplex systems

We consider hypercomplex systems $\mathfrak{o}$ relative to a commutative field P. By convention, only P-modules are counted as ideals in $\mathfrak{o}$. Hence the maximal and minimal conditions hold for left and right ideals, and the entire ring theory of Chapter II applies.

In what follows, a representation of the hypercomplex system $\mathfrak{o}$ always means a representation in the field P, and more precisely one satisfying: from $a \mapsto A$ follows

$$a\rho \mapsto A\rho \qquad (\rho \in P),$$

that is, module homomorphy relative to P. For representation modules this says

$$a\rho \cdot m = a(m\rho).$$

The results of §19 therefore apply. All irreducible representations are mediated by simple left ideals of

$$\mathfrak{o}_0 = \mathfrak{o}/\mathfrak{c},$$

where $\mathfrak{c}$ is the radical. Thus there are as many inequivalent irreducible representations as there are two-sided simple summands $\mathfrak{a}$ in a decomposition

$$\mathfrak{o}_0 = \mathfrak{a}_1 + \cdots + \mathfrak{a}_s.$$

To determine explicitly the representation defined by such a left ideal, first pass from the elements of $\mathfrak{o}$ to their residue classes in $\mathfrak{o}_0$. Multiplication by an element of P is an isomorphism for all ideals of $\mathfrak{o}_0$; hence P is isomorphic to a subfield $e^{(\nu)}P$ of the automorphism division ring K_ν of every simple left ideal. The representation over P mediated by a simple left ideal $\mathfrak{l}_\nu$ is obtained by first studying the representation over this subfield $e^{(\nu)}P$, and then transporting it to P through the isomorphism

$e^{(\nu)}P \simeq P$. The division ring K_ν has finite rank over P; thus we are dealing with representation in a finite-degree subfield of the automorphism division ring of $\mathfrak{l}_\nu$.

To find the representation, write each element c of $\mathfrak{o}_0$ according to its two-sided components,

$$c = c_1 + \cdots + c_s.$$

Only the matrix of c_ν is then relevant, since the other c_i annihilate $\mathfrak{l}_\nu$ and are represented by zero matrices. Write c_ν with respect to the matrix units $c_{ij}^{(\nu)}$ of $\mathfrak{a}_\nu$:

$$c_\nu = \sum_{i,j} c_{ij}^{(\nu)} \alpha_{ij}^{(\nu)}.$$

Then replace each entry $\alpha_{ij}^{(\nu)}$ by the matrix which represents it in the representation of K_ν over P.

The division ring K_ν has finite rank over P. Every element $x \in K_\nu$ satisfies an equation $f(x) = 0$ with coefficients in $e^{(\nu)}P$, since its powers are linearly dependent. If P is algebraically closed, this equation splits into linear factors; because K_ν is a division ring, x is already a root of one of those linear factors, and so $x \in e^{(\nu)}P$. Thus $K_\nu = e^{(\nu)}P$. In this case the matrices $(\alpha_{ij}^{(\nu)})$ themselves are the irreducible representations.

Combining this observation with the end of §19 gives Burnside's theorem in the following form.

Let P be algebraically closed and commutatively connected with a ring $\mathfrak{o}$. Then every irreducible representation of degree n over P contains exactly n^2 linearly independent matrices.

This immediately implies the usual formulation: a system of n by n matrices over P, closed under multiplication, is irreducible if and only if there is no non-trivial homogeneous linear relation

$$\sum \alpha_{ij} x_{ij} = 0$$

with coefficients in P satisfied by the entries x_{ij} of every matrix in the system. For by adjoining all linear combinations one obtains a ring and a P-module, and may regard this ring as its own representation.

Indeed, the absolute multiplier domain is isomorphic to a two-sided simple completely reducible ring with unit, hence to a full matrix ring over the automorphism division ring of its left ideals. Algebraic closedness makes that division ring equal to P. An irreducible representation is generated by a simple left ideal. If this left ideal has rank m, and the representation has degree n, then $m = n$, and there are exactly n^2 linearly independent elements in the full matrix ring.

Similarly one obtains the generalized Burnside theorem. Under the same hypotheses, if D is a completely reducible representation which decomposes into pairwise inequivalent constituents of degrees

$$n_1, n_2, \ldots, n_s,$$

then D has rank

$$n_1^2 + n_2^2 + \cdots + n_s^2$$

over P.

If $\mathfrak{o}$ is a hypercomplex system without radical over an arbitrary field, then $\mathfrak{o}$ is completely reducible and has a unit element. It follows from §§17 and 18 that every representation of $\mathfrak{o}$ is completely reducible.

Conversely, every completely reducible representation of a ring $\mathfrak{o}$ commutatively connected with P has as absolute multiplier domain a hypercomplex system without radical. Indeed, the absolute multiplier domain is isomorphic to a ring and P-module of matrices over P, hence is a hypercomplex system. The elements of its radical act as zero in every irreducible constituent and hence in the whole representation; therefore the radical is zero.

The **regular representation** of a hypercomplex system $\mathfrak{o}$ is the representation mediated by the unit ideal $\mathfrak{o}$ itself. For a group ring one chooses specifically the basis consisting of the group elements. Since all left ideals of $\mathfrak{o}/\mathfrak{c}$ occur as composition factors in $\mathfrak{o}$, all irreducible representations already occur on the diagonal of the regular representation, after reducing the regular representation by a composition series as in §16.

A representation is known once the representing matrices of the basis elements $u_1, \ldots, u_h$ are known. If the system is a group ring, so that the u_i are group elements, every homomorphic matrix representation of the group induces a representation of the group ring. Thus the representation problem for groups is a special case of the representation problem for hypercomplex systems.

§21. Extension of the ground field. Representations of the center

Let

$$\mathfrak{o} = a_1 P + \cdots + a_h P$$

be a hypercomplex system, and let Ω be an extension field of P. One forms

$$\mathfrak{o}\Omega = a_1 \Omega + \cdots + a_h \Omega$$

with the old multiplication rules for the basis elements. This is again hypercomplex over Ω.

Every representation of $\mathfrak{o}$ gives a representation of $\mathfrak{o}\Omega$, since the representation of all elements is known as soon as the representations of the basis elements are known.

An ideal, a representation module, or the corresponding representation is called **absolutely irreducible** if it remains irreducible after passage to the algebraic closure.

If $\mathfrak{o}\Omega$ has no radical, then neither has $\mathfrak{o}$: a nilpotent ideal of $\mathfrak{o}$ would give a nilpotent extended ideal in $\mathfrak{o}\Omega$. The converse is false, as will appear below.

The center of $\mathfrak{o}\Omega$ is the extended center $Z\Omega$. If $\mathfrak{o}$ is completely reducible, then

$$Z = Z_1 + \cdots + Z_s$$

is a direct sum of fields. If, after passage to Ω, the algebra $\mathfrak{o}\Omega$ remains completely reducible, so that no nilpotent ideal appears, then the new center

$$Z\Omega = Z_1\Omega + \cdots + Z_s\Omega$$

is again a direct sum of fields. If Ω is algebraically closed, these fields have rank one over Ω.

For the representations of the center the following statements hold.

Every irreducible representation of a commutative hypercomplex system Z over an algebraically closed field Ω has degree one. Equivalently, irreducible representations are the same as homomorphisms

$$Z \to \Omega.$$

Indeed, every irreducible representation of Z gives one of $Z\Omega$, and hence of $Z\Omega/\mathfrak{c}$, where $\mathfrak{c}$ is the radical. This quotient is commutative and without radical, hence a direct sum of fields. Over an algebraically closed field these fields are of first degree, and they generate all irreducible representations. Since equivalent one-dimensional representations are necessarily equal, the number of distinct homomorphisms $Z \to \Omega$ is the rank of $Z\Omega/\mathfrak{c}$.

For the special case in which Z is a field over P, this says: Z is completely reducible, i.e. has no radical after scalar extension, if and only if Z is an extension of the first kind over P. For a field the homomorphisms are isomorphisms. Their number is the rank of $Z\Omega/\mathfrak{c}$, and it equals the field degree precisely when $\mathfrak{c} = 0$.

If

$$Z = Z_1 + \cdots + Z_s$$

is completely reducible, then under every irreducible representation exactly one of the fields Z_i is mapped isomorphically, and the others are represented by zero.

Applying these statements to hypercomplex systems gives first, for systems without radical: if $\mathfrak{o}\Omega$, and therefore $\mathfrak{o}$, is without radical, then in every absolutely irreducible representation of $\mathfrak{o}$ the central elements z are represented by scalar diagonal matrices

$$\xi E,$$

and $z \mapsto \xi$ is a one-dimensional representation of the center. The correspondence between absolutely irreducible representation classes of $\mathfrak{o}$ and those of Z is one-to-one. Hence the number of such classes equals the rank of the center.

For the decomposition into two-sided indecomposable ideals

$$\mathfrak{o}\Omega = \mathfrak{a}_1 + \cdots + \mathfrak{a}_s$$

corresponds to a decomposition of the center into fields

$$Z\Omega = Z_1 + \cdots + Z_s.$$

In an irreducible representation one $\mathfrak{a}_i$ is represented isomorphically, the others by zero; this two-sided summand determines the representation class. The summand $\mathfrak{a}_i$ is represented by a full matrix ring, and the center of such a ring consists of scalar diagonal matrices. Thus the representation of the center is a homomorphism onto the corresponding scalar field, and this proves the assertion.

The following consequence is useful. If the center has a component Z_i which is an extension of the second kind over P, then $\mathfrak{o}\Omega$ certainly has a radical, since $Z_i\Omega$ already has one.

§22. Application to abelian groups

Let

$$G = G_1 \times G_2 \times \cdots \times G_s$$

be a finite abelian group, decomposed into cyclic groups G_i of orders h_i. Let

$$G_i = \{1, a_i, a_i^2, \ldots, a_i^{h_i - 1}\}.$$

Let P be a field whose characteristic does not divide the group order

$$h = h_1 h_2 \cdots h_s.$$

The group ring Z consists of all sums

$$\sum c_{i_1 \ldots i_s} a_1^{i_1} \cdots a_s^{i_s}, \qquad c_{i_1 \ldots i_s} \in P.$$

It is a homomorphic image of the polynomial ring $P[x_1, \ldots, x_s]$ by the substitution $x_i \mapsto a_i$. Hence

$$Z \simeq P[x_1, \ldots, x_s]/\mathfrak{m},$$

where

$$\mathfrak{m} = (x_1^{h_1} - 1, \ldots, x_s^{h_s} - 1).$$

In an extension field Ω in which all these polynomials split, each $x_i^{h_i} - 1$ decomposes into distinct linear factors, since the characteristic does not divide h_i. Thus $\mathfrak{m}$ decomposes as the intersection of distinct prime ideals

$$(x_1 - \xi_1, \ldots, x_s - \xi_s),$$

where each ξ_i runs through the h_i-th roots of unity. The direct-sum decomposition corresponding to this intersection gives one field for each system $(\xi_1, \dots, \xi_s)$. The associated representations are

$$a_i \mapsto \xi_i, \qquad a_1^{i_1} \cdots a_s^{i_s} \mapsto \xi_1^{i_1} \cdots \xi_s^{i_s}.$$

These are the characters. Their number is

$$h_1 h_2 \cdots h_s = h,$$

as the general theory requires.

§23. Determinant of a hypercomplex system

Let

$$\mathfrak{o} = a_1 P + \cdots + a_h P$$

be a hypercomplex system. Adjoin to P independent variables $x_1, \dots, x_h$ and form

$$\mathfrak{o}^* = a_1 P(x_1, \dots, x_h) + \cdots + a_h P(x_1, \dots, x_h)$$

with the old multiplication rules; the variables commute with the a_i. Thus the rules in $\mathfrak{o}^*$ are already determined.

In $\mathfrak{o}^*$ lies the **general element** of $\mathfrak{o}$,

$$w = a_1 x_1 + \cdots + a_h x_h.$$

If $a_i \mapsto A_i$ in a representation, then w is represented by

$$W = A_1 x_1 + \cdots + A_h x_h.$$

The matrix W is the system matrix attached to the representation; when the a_i are group elements and $\mathfrak{o}$ is the group ring, it is the group matrix. For the regular representation one has the regular system matrix.

The entries of W are linear forms in the x_i. The determinant $|W|$ is therefore a homogeneous form of degree n when the representation has degree n. In particular, the regular system determinant has degree h.

The system determinant does not change under passage to an equivalent representation, because

$$|PWP^{-1}| = |P|\,|W|\,|P^{-1}| = |W|.$$

If one changes from the basis $(a_1, \ldots, a_h)$ to a new basis $(b_1, \ldots, b_h)$ and writes

$$w = \sum_i b_i y_i,$$

then the new determinant is obtained from the old one by an invertible linear substitution

$$x_i = \sum_j \gamma_{ij} y_j.$$

If a composition series of the representation module is given, then, for a suitable choice of basis, the matrix W is upper triangular in blocks:

$$\begin{pmatrix} W_1 & * & \cdots & * \\ 0 & W_2 & \cdots & * \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & W_r \end{pmatrix}.$$

The determinants $|W_i|$ which occur in an arbitrary representation are among those occurring in the regular representation of $\mathfrak{o}$, indeed already in that of $\mathfrak{o}/\mathfrak{c}$, where $\mathfrak{c}$ is the radical.

Assume now that P is algebraically closed. Then the determinant belonging to an irreducible representation is an irreducible polynomial in the x_i, and inequivalent representations give different irreducible factors.

It is enough to work in the radical-free ring $\mathfrak{o}/\mathfrak{c}$. There we use matrix units $c_{ij}^{(\nu)}$ as a basis; the general element has the form

$$w = \sum_{\nu,i,j} c_{ij}^{(\nu)} x_{ij}^{(\nu)}.$$

The matrices of the irreducible representations are precisely

$$W_\nu = (x_{ij}^{(\nu)}).$$

The determinants

$$|W_\nu| = |x_{ij}^{(\nu)}|$$

are the familiar irreducible determinant forms and are plainly distinct.

To compute the $|W_\nu|$, one may factor the regular system matrix of $\mathfrak{o}/\mathfrak{c}$. Each irreducible factor appears with multiplicity equal to the degree of the corresponding irreducible representation, since the corresponding left ideal occurs that many times in a composition series.

One may also start from the regular system matrix of $\mathfrak{o}$ itself. Then each irreducible factor appears with a higher multiplicity, namely as often as the corresponding simple left ideal occurs as composition factor in the left ideals of $\mathfrak{o}$. If $\mathfrak{o}$ is instead regarded as right ideal, one gets a second regular system matrix, Frobenius's antistrophic matrix. It contains the same irreducible factors -

the system determinants of all irreducible representations - possibly with different exponents.

The system determinant of a commutative system splits into linear factors, because all irreducible representations have degree one. These linear factors are themselves the irreducible representations and, for abelian groups, the characters. This was Dedekind's starting point in the study of the group determinant of non-abelian groups.

§24. Traces and characters

If, in a representation of a hypercomplex system $\mathfrak{o}$, the element a is assigned the matrix A, set

$$\mathrm{Sp}(a) = \mathrm{Sp}\, A.$$

Traces are linear functions:

$$\mathrm{Sp}(c + d) = \mathrm{Sp}(c) + \mathrm{Sp}(d), \qquad \mathrm{Sp}(\alpha c) = \alpha\, \mathrm{Sp}(c).$$

Equivalent representations have the same traces.

The trace in a reducible representation is the sum of the traces in the representations mediated by the composition factors, because the representing matrix is block triangular and its trace is the sum of the traces of the diagonal blocks.

The **principal trace** is the trace in the regular representation. The **reduced trace** is the sum of the traces in the distinct irreducible representations.

If c belongs to the maximal nilpotent ideal $\mathfrak{c}$, then

$$\mathrm{Sp}(c) = 0$$

for every representation. It suffices to consider the representations mediated by the simple left ideals of $\mathfrak{o}/\mathfrak{c}$, since in every other representation only these composition factors occur and trace is additive. But $\mathfrak{c}$ annihilates all elements of $\mathfrak{o}/\mathfrak{c}$; hence every $c \in \mathfrak{c}$ is represented by the zero matrix.

Let $\mathfrak{o}$ be two-sided simple and let P be algebraically closed; thus $\mathfrak{o}$ is a matrix ring

$$\mathfrak{o} = \sum c_{ij} P.$$

In the irreducible representation, the element

$$a = \sum c_{ij} \alpha_{ij}$$

is assigned the matrix (α_{ij}). Hence

$$\mathrm{Sp}_{\mathrm{red}}(a) = \sum_i \alpha_{ii},$$

while the principal trace is

$$\mathrm{Sp}_{\mathrm{pr}}(a) = n \sum_i \alpha_{ii}.$$

In particular,

$$\mathrm{Sp}_{\mathrm{red}}(c_{ij}) = 0 \quad (i \neq j), \qquad \mathrm{Sp}_{\mathrm{red}}(c_{ii}) = 1,$$

and

$$\mathrm{Sp}_{\mathrm{pr}}(c_{ii}) = n.$$

Passing from P to an algebraically closed field Ω does not change the principal trace, since it may be computed from the same basis. This is not true for the reduced trace.

The traces of the elements a of the radical-free system $\mathfrak{o}$ in the absolutely irreducible representations are called **characters** and are denoted by

$$\chi(a),$$

or, if the representation is to be specified, by

$$\chi^{(\nu)}(a).$$

In an irreducible representation of degree n_ν, the central elements, by §21, are represented by scalar matrices

$$\xi^{(\nu)}E,$$

where $z \mapsto \xi^{(\nu)}$ is a one-dimensional representation of the center, i.e. a homomorphism of the center into Ω. Hence the trace has value $n_\nu \xi^{(\nu)}$. Thus the homomorphisms Θ of the center and the characters χ are connected by

$$\chi(z) = n_\nu \Theta(z).$$

In the commutative case $n_\nu = 1$, and the characters themselves are the homomorphisms.

If the algebraically closed field Ω has characteristic zero, as will be assumed below, one may divide by n_ν:

$$\Theta(z) = \frac{\chi(z)}{n_\nu}.$$

The homomorphism property of Θ becomes

$$\frac{\chi(z)}{n_\nu} + \frac{\chi(z')}{n_\nu} = \frac{\chi(z+z')}{n_\nu},$$

and

$$\frac{\chi(z)}{n_\nu}\frac{\chi(z')}{n_\nu} = \frac{\chi(zz')}{n_\nu}.$$

A representation class is already uniquely determined by the traces of its matrices. To know all traces it suffices, of course, to know the traces of the basis elements in the given representation. If the representation module is R, it is determined by the multiplicities with which the irreducible modules M_ν occur as direct summands. If this multiplicity is p_ν, then the trace of the idempotent $e^{(\nu)}$ in the representation equals

$$p_\nu n_\nu.$$

Thus

$$p_\nu = \frac{\mathrm{Sp}(e^{(\nu)})}{n_\nu}.$$

§25. Discriminants

Let

$$\mathfrak{o} = a_1 P + \cdots + a_h P$$

be a hypercomplex system.

The **discriminant matrix** is the matrix whose entries are the principal traces

$$\mathrm{Sp}_{\mathrm{pr}}(a_i a_j).$$

The **reduced discriminant matrix** is the analogous matrix formed from reduced traces after passing to the algebraically closed field. The determinants of these matrices are the discriminant and reduced discriminant.

The discriminant remains the same under extension of the ground field. Under passage to another basis it is multiplied by the square of the transformation determinant.

Indeed, suppose

$$b_j = \sum_i a_i p_{ij}.$$

Then

$$(\mathrm{Sp}(a_i a_j)) = (\mathrm{Sp}(a_i b_j))P,$$

and similarly, using the transposed matrix,

$$(\mathrm{Sp}(a_i b_j)) = P^t(\mathrm{Sp}(b_i b_j)).$$

Thus

$$(\mathrm{Sp}(a_i a_j)) = P^t(\mathrm{Sp}(b_i b_j))P,$$

and taking determinants proves the assertion. The determinant is therefore determined only up to a square factor from P, but its vanishing or non-vanishing is invariant. The same calculation also shows that, up to a non-zero factor, one may compute the discriminant from two different bases as

$$\left|\mathrm{Sp}(a_i b_j)\right|.$$

The discriminant vanishes if $\mathfrak{o}$ has a nilpotent ideal $\mathfrak{c}$. Choose a basis

$$c_1, \ldots, c_r, d_1, \ldots, d_s$$

so that $c_1, \ldots, c_r$ is a basis of $\mathfrak{c}$. Since products $c_i c_j$ are nilpotent, and the trace of nilpotent elements is zero in every representation, the corresponding block of the discriminant matrix is zero, forcing the determinant to vanish. The same holds for the reduced discriminant.

Consequently, the discriminant also vanishes if a nilpotent ideal appears only after passing to the algebraic closure.

If

$$\mathfrak{o} = \mathfrak{a}_1 + \cdots + \mathfrak{a}_s$$

is a direct two-sided decomposition, and if $M, M_1, \ldots, M_s$ are the discriminant matrices of $\mathfrak{o}, \mathfrak{a}_1, \ldots, \mathfrak{a}_s$, then M is block diagonal with blocks M_i, and hence

$$|M| = |M_1| \cdots |M_s|.$$

Indeed, choose a basis of $\mathfrak{o}$ composed of bases of the $\mathfrak{a}_i$. If $a \in \mathfrak{a}_i$, then its principal trace computed in $\mathfrak{o}$ is the same as its principal trace computed in $\mathfrak{a}_i$. Products from different components vanish, so the off-diagonal trace blocks are zero. The same reasoning applies to the reduced discriminant.

For the full matrix ring

$$\sum c_{ij} P$$

one can compute the discriminant by taking two bases consisting of the matrix units, once ordered by first indices and once by second indices. The product table shows that

$$\mathrm{Sp}(c_{ij} c_{kl})$$

vanishes except in the matching cases. The resulting determinant is

$$D_{\text{red}} = 1$$

for reduced traces, and

$$D = n^{n^2}$$

for principal traces.

Thus, if $\mathfrak{o}$ decomposes over the algebraic closure into matrix rings of degrees

$$n_1, \ldots, n_s,$$

then

$$D = (n_1^{n_1^2} \cdots n_s^{n_s^2})\, e$$

up to a non-zero square factor, whereas the reduced discriminant is non-zero. In particular, the reduced discriminant is non-zero for radical-free systems, and the principal discriminant is non-zero precisely when none of the n_i is divisible by the characteristic of the field.

Hence non-vanishing of the reduced discriminant is necessary and sufficient for the absence of radical after algebraic closure of the field P. In characteristic zero, non-vanishing of the principal discriminant is likewise necessary and sufficient for the absence of radical after algebraic closure.

§26. The group ring

Let $a_1, \ldots, a_h$ be the elements of a finite group, and form the group ring over a field whose characteristic does not divide h. Then the discriminant D is non-zero; hence the group ring is radical-free.

For the principal trace in the regular representation satisfies

$$\mathrm{Sp}(1) = h,$$

and

$$\mathrm{Sp}(a_i) = 0$$

for $a_i \neq 1$. In the regular representation the identity is represented by a diagonal matrix with h diagonal entries equal to 1. If $a_i \neq 1$, then multiplication by a_i permutes the group elements without fixed points; the corresponding matrix therefore has zeros on the main diagonal, so its trace is zero.

Take the two bases

$$a_1, \ldots, a_h$$

and

$$a_1^{-1}, \ldots, a_h^{-1}.$$

The matrix

$$(\mathrm{Sp}(a_i a_j^{-1}))$$

is diagonal with diagonal entries h. Therefore

$$D = h^h \neq 0.$$

The **class** of a group element is the sum, formed in the group ring,

$$K_a = \sum_s s^{-1} a s,$$

where the sum ranges over the distinct conjugates. The class sums K_a are central, since they commute with all group elements. They generate the center: if an element $\sum \alpha_i a_i$ of the group ring commutes with all s, then

$$\sum_i \alpha_i a_i = s^{-1}\left(\sum_i \alpha_i a_i\right) s$$

for every s, so the coefficients are constant on conjugacy classes.

Thus the rank of the center is the number of conjugacy classes. Consequently the number of absolutely irreducible representations is equal to the number of classes of conjugate group elements.

The matrices A and $S^{-1}AS$ have the same trace; hence conjugate group elements have the same character. Taking the trace of the class sum gives

$$\chi(K_a) = h_a \chi(a),$$

where h_a is the number of elements in the conjugacy class of a. In words: the character of a group element is the character of its class sum divided by the number of elements of that class.

35. On Maximal Domains of Integral Functions

The following investigations are closely connected with Hilbert's problem on relatively integral functions. They go back to an occasional question of E. Hecke, who needed a special case as an auxiliary lemma and handled it by direct calculation.

The question is this. By a maximal domain of integral functions in the variables $x_1, \ldots, x_n$ - polynomials, power series, or quotients of such - inside a given domain $\mathfrak{B}$, I mean a domain which can no longer be enlarged by adjoining integral denominators. Thus, whenever $a\,g(x)$ belongs to it, with $g(x) \in \mathfrak{B}$ and a a non-zero rational integer, then $g(x)$ itself must already belong to it.

Every integral domain $\mathfrak{A}$ in $\mathfrak{B}$ generates uniquely such a maximal domain $M(\mathfrak{A})$. It consists of all functions $g(x)$ in $\mathfrak{B}$ for which there is at least one non-zero integer a such that

$$ag(x) \in \mathfrak{A}.$$

The question is what can be said about the denominators a of this maximal domain when the original integral domain $\mathfrak{A}$ was finite, that is, when it consists of all integral polynomials in finitely many functions

$$f_1(x), \ldots, f_r(x)$$

from $\mathfrak{B}$.

For $\mathfrak{B}$ one may take all integral polynomials or power series in $x_1, \ldots, x_n$, or else those quotients which contain no denominators except those already occurring among the $f_i(x)$, and their powers. I show - under the additional hypothesis, in the case of power series or quotients of power series, that only algebraic relations occur among the $f_i(x)$ - that the integers a may always be chosen so that only finitely many different prime numbers occur in them. The powers of those primes, however, need not be bounded.

This last point is immediate. If $ag(x)$ belongs to $\mathfrak{A}$ but $a^\rho g(x)$ does not belong to $\mathfrak{A}$ for any fixed bound on ρ, then all powers of a may occur. Thus the reasonable boundedness question is different: can one give, in general, an infinite system of generators of $M(\mathfrak{A})$ in which the exponents of the primes in the corresponding denominators remain bounded? Since only finitely many primes are involved, this is equivalent, by known arguments, to the finiteness of $M(\mathfrak{A})$ itself. It is therefore Hilbert's problem on relatively integral functions in its integral form. The methods used here do not decide that question.

The proof that only finitely many different primes p occur in the denominators rests on the following idea. To every such prime there corresponds at least one relation modulo p among the functions $f_i(x)$ generating $\mathfrak{A}$, a relation with integral coefficients which is not identically satisfied in the variables x before reduction. In another language, let $\mathfrak{o}$ be the polynomial ring over the integers in indeterminates $u_1, \ldots, u_r$. The assignment

$$u_i \longmapsto f_i(x)$$

makes $\mathfrak{A}$ a homomorphic image of $\mathfrak{o}$; the homomorphism is defined by a prime ideal $\mathfrak{a}$ of $\mathfrak{o}$, so that

$$\mathfrak{A} \simeq \mathfrak{o}/\mathfrak{a}.$$

After reduction modulo p, $\mathfrak{B}_p$ is likewise a homomorphic image of the reduced polynomial ring $\mathfrak{o}_p$ except for the finitely many primes already appearing in the denominators of the $f_i(x)$. The new homomorphism is defined by a prime ideal $\mathfrak{c}_p$ of $\mathfrak{o}_p$ containing $\mathfrak{a}_p$.

A prime p occurs in the denominator of $M(\mathfrak{A})$ if and only if $\mathfrak{c}_p$ is a proper over-ideal of $\mathfrak{a}_p$. This can happen only if either the transcendence degree of $\mathfrak{B}_p$ drops in comparison with that of $\mathfrak{A}$, or if $\mathfrak{a}_p$ ceases to be a prime ideal. Both events occur only for finitely many primes. The first is controlled by functional determinants modulo p; the second by the fact that $\mathfrak{a}$ is an absolute prime ideal and can lose primality only modulo finitely many primes.

All arguments remain valid, with minor modifications, if the rational integers are replaced by the integers of a finite algebraic number field and the rational primes by the prime ideals of that field.

§1. Functional determinants over coefficient fields of characteristic p

Let

$$f_1(x_1,\ldots,x_n),\ldots,f_r(x_1,\ldots,x_n)$$

be rational functions, or more generally formally defined power series or quotients of such, with coefficients in a field P of characteristic p. Assume that the derivatives $\partial f_i/\partial x_j$ are formally defined and satisfy the usual rules of calculation. Let Ω be an algebraically closed extension field of P.

Then the following statement holds as in characteristic zero. If there is an algebraic dependence among the $f_i(x)$ with coefficients in Ω, then the rank of the functional matrix

$$\left(\frac{\partial f_i}{\partial x_j}\right)$$

is smaller than the number of algebraically independent functions under consideration, unless the dependence is purely inseparable. More precisely, if the functions are algebraically dependent and $G(u_1,\ldots,u_r)$ is a non-zero polynomial relation of least degree, then

$$G(f_1,\ldots,f_r)=0.$$

Differentiation gives

$$\sum_i \frac{\partial G}{\partial u_i}(f)\frac{\partial f_i}{\partial x_j}=0$$

for every j. If the functional matrix had full rank, all partial derivatives $\partial G/\partial u_i$ would vanish. In characteristic p, this means that G is a polynomial in the p-th powers,

$$G(u)=H(u_1^p,\ldots,u_r^p).$$

Since Ω is algebraically closed, this contradicts the choice of G as an irreducible polynomial of least degree. Thus the rank must drop.

The consequence needed below is the following. Let

$$F_1(x), \ldots, F_t(x)$$

be integral functions - polynomials, power series, or quotients of such - whose functional matrix has rank exactly t. Choose a prime p which occurs neither in the denominators of the $F_i(x)$ nor in all t-rowed determinants of the functional matrix. Then the reduced functions

$$f_1(x), \ldots, f_t(x)$$

modulo p remain algebraically independent.

Indeed, their functional matrix is obtained by reducing the entries of the original functional matrix modulo p. By the choice of p, the rank remains t. Algebraic dependence would force the rank to be smaller, by the preceding argument.

§2. Formulation of the theorem

Let $\mathfrak{A}$ be an integral domain of integral functions in $x_1, \ldots, x_n$: polynomials, power series, or quotients of such, with rational integral coefficients. In the case of power series one may understand formal power series or convergent power series; only the fact that the functions form an integral domain is used.

Let $\mathfrak{Q}$ be the quotient field of $\mathfrak{A}$, and let $\mathfrak{B}$ be an integral domain between $\mathfrak{A}$ and $\mathfrak{Q}$. The domain $\mathfrak{M}$ is called maximal in $\mathfrak{B}$ if the following condition holds: whenever $a \neq 0$ is an integer, $g(x) \in \mathfrak{B}$, and $ag(x) \in \mathfrak{M}$, then $g(x) \in \mathfrak{M}$.

The maximal domain generated by $\mathfrak{A}$ in $\mathfrak{B}$ exists uniquely. It is the intersection of all maximal domains in $\mathfrak{B}$ containing $\mathfrak{A}$, and it consists exactly of those $g(x) \in \mathfrak{B}$ for which $ag(x) \in \mathfrak{A}$ for at least one non-zero integer a.

Assume now that $\mathfrak{A}$ is finite and has a unit element. Let

$$f_1(x), \ldots, f_r(x)$$

be an integral basis, so that $\mathfrak{A}$ consists of all integral polynomials in these functions. Let $\mathfrak{B}$ consist of the allowed polynomials, power series, or quotients, with no denominators except those appearing among the finitely many $f_i(x)$.

In the case of power series or quotients one adds the following condition. If t is the rank of the functional matrix of the $f_i(x)$, and if, after renumbering, $f_1, \ldots, f_t$ already have a functional matrix of rank t, then the quotient field $\mathfrak{Q}$ is to be a finite algebraic extension of

$$\mathbb{Q}(f_1, \ldots, f_t).$$

Equivalently, the remaining $f_{t+1}, \ldots, f_r$ are algebraic over the field generated by $f_1, \ldots, f_t$.

Let $\mathfrak{o} = \mathbb{Z}[u_1, \ldots, u_r]$. Then $\mathfrak{A}$ is a homomorphic image of $\mathfrak{o}$ under $u_i \mapsto f_i(x)$, hence

$$\mathfrak{A} \simeq \mathfrak{o}/\mathfrak{a}$$

for a prime ideal $\mathfrak{a}$. The theorem says that, except for finitely many rational primes p, reduction modulo p gives no new denominator obstruction: if a is not divisible by any of those exceptional primes and $ag(x)$ lies in $\mathfrak{A}$, then $g(x)$ already lies in $\mathfrak{A}$.

Equivalently, only finitely many primes can occur in the denominators required to pass from $\mathfrak{A}$ to its maximal domain.

§3. Proof of the theorem

The first auxiliary assertion is that the prime ideal $\mathfrak{a}$ which defines the homomorphism from $\mathfrak{o}$ to $\mathfrak{A}$ is absolutely prime, under the stated hypotheses. In other words, after extending the coefficient field to an algebraic closure, the corresponding ideal remains prime. This is a standard consequence of the irreducibility of the system $f_1, \ldots, f_t$ and of the algebraic dependence of the remaining functions over it.

Therefore $\mathfrak{a}$ can cease to be prime after reduction modulo p for at most finitely many primes. This is one part of the exceptional set. A second finite exceptional set is formed by the primes in the denominators of the $f_i(x)$. A third is formed by the primes at which all suitable t-rowed functional determinants vanish modulo p. Outside these finitely many primes, three facts hold simultaneously:

1. the reduction of the functions $f_i(x)$ is defined;
2. the reduced ideal $\mathfrak{a}_p$ is still prime;
3. the reduced functions $f_1, \ldots, f_t$ remain algebraically independent.

Choose p outside the exceptional set. The homomorphism from the reduced polynomial ring $\mathfrak{o}_p$ to $\mathfrak{B}_p$ is defined by a prime ideal $\mathfrak{c}_p$ containing $\mathfrak{a}_p$. It remains to show that in fact

$$\mathfrak{c}_p = \mathfrak{a}_p.$$

For this it is enough to compare transcendence degrees. The residue field of $\mathfrak{c}_p$ is the quotient field of the reduced image domain, so its transcendence degree is at least t. The residue field of $\mathfrak{a}_p$ also has transcendence degree exactly t, because the classes of $u_1, \ldots, u_t$ remain algebraically independent and the remaining classes are algebraic over them.

To see the latter point explicitly, use the algebraic relations in characteristic zero. The ideal $\mathfrak{a}$ contains primitive integral polynomials

$$G_{t+1}(u_1, \ldots, u_t, u_{t+1}), \ldots, G_r(u_1, \ldots, u_t, u_r)$$

expressing the algebraic dependence of the remaining generators. For all but finitely many primes these polynomials do not vanish after reduction and still involve the corresponding variable u_j genuinely. Hence the classes of $u_{t+1}, \ldots, u_r$ are algebraic over the classes of $u_1, \ldots, u_t$ modulo $\mathfrak{a}_p$.

Thus $\mathfrak{a}_p$ and $\mathfrak{c}_p$ are prime ideals of the same transcendence degree, and $\mathfrak{c}_p$ contains $\mathfrak{a}_p$. A prime ideal has no proper prime over-ideal of the same transcendence degree. Therefore $\mathfrak{c}_p = \mathfrak{a}_p$.

This proves that no such prime p can occur in a denominator. The theorem follows.

§4. Algebraic integer coefficient domains

The same reasoning applies if the rational integers are replaced by the ring of integers of a finite algebraic number field K, and rational primes by prime ideals of K.

The determinant argument and the formulation of the theorem are unchanged. In the proof, however, one must add to the exceptional prime ideals those which divide the finitely many algebraic integers occurring as contents of the polynomials expressing the algebraic dependences; these polynomials can no longer all be assumed primitive over $\mathbb{Z}$ in the same naive sense.

The final formulation is as follows. Let a be an integer of K not divisible by any of the finitely many exceptional prime ideals. If $ag(x)$ lies in the original domain $\mathfrak{A}$ and $g(x) \in \mathfrak{B}$, then $g(x)$ lies in the maximal domain generated by $\mathfrak{A}$ without needing any of the non-exceptional prime ideals as denominators.

Equivalently, after localizing away from the exceptional prime ideals, the same argument as above shows that the relevant reduced homomorphisms have the same prime ideal kernel. If

$$(a) = \mathfrak{p}_1^{e_1} \cdots \mathfrak{p}_s^{e_s}$$

is the prime-ideal factorization and none of the $\mathfrak{p}_i$ is exceptional, the localization makes these prime ideals principal and leaves the ideal decomposition unchanged. The preceding proof then applies word for word.

If the exceptional prime ideals are generated in the localized ring by elements

$$\lambda_1, \ldots, \lambda_q,$$

then every needed denominator may be chosen as a product of powers of these finitely many λ_i. Returning from the localized ring to the original ring of integers gives the asserted finite-prime statement.

36. Ideal Differentiation and the Different

It is shown how the different of an algebraic number field may be understood as a differential quotient of a defining ideal. The agreement of this definition with the usual one follows from structural theorems which are interesting in themselves and which, in an analogous sense, generalize Lagrange's interpolation formula. A full account is to appear in the *Mathematische Annalen*.

37. Normal Basis for Fields Without Higher Ramification

By a normal basis of a Galois number field K/k one means a basis of the maximal order $\mathfrak{O}$ of K over the maximal order $\mathfrak{o}$ of k which consists of conjugate elements. It is known that a necessary condition for the existence of such a basis is the absence of higher ramification groups in K/k. The same remains true locally: after passing to the $\mathfrak{p}$-adic extension $k_\mathfrak{p}$ of k and the corresponding extension $K_\mathfrak{p}$ of K.

I prove the converse in the following form. For every prime $\mathfrak{p}$ whose residue characteristic does not divide the degree of K/k, there exists a normal basis. In particular, if K/k has no higher ramification, then at every ramified prime there is a normal basis; the discriminant at such a prime becomes the square of a group determinant.

This last assertion must be supplemented. Contrary to a conjecture of E. Artin, the passage from this factorization to Artin conductors still requires further ideal-theoretic considerations. The decomposition of the group determinant according to irreducible characters gives generalized root numbers. A suitable collection of these gives a factorization in the ground field enlarged by the character values. In the simplest cases these new factors can be identified with Artin's conductors by means of the ideal-theoretic factorization of the root numbers.

Even in the general case, where no normal basis exists, a decomposition into generalized root numbers is always possible by means of a composition series as Galois modules. This again gives a factorization in the enlarged ground field, and again ideal-theoretic questions enter.

The proof of the existence of the normal basis under the hypotheses stated above rests on this point: the module $\mathfrak{O}/\mathfrak{o}$, regarded as a Galois module - that is, as an $\mathfrak{o}$-module with the substitutions of the Galois group as operator domain - becomes operator-isomorphic to a one-sided ideal of the integral group ring enlarged by $\mathfrak{o}$. At all primes with only ordinary ramification this group ring is a maximal order. Ideals in maximal orders of p-adic semisimple systems are principal. Hence the corresponding Galois module is generated by one element under the substitutions of the group, which is exactly the assertion that there is a normal basis.

At primes with higher ramification the integral group ring passes to a non-maximal order, and the ideal attached to $\mathfrak{O}/\mathfrak{o}$ passes to a non-principal ideal. The same reasoning applies to function fields of one variable, provided that the characteristic of the constant field does not divide the degree of K/k.

§1. The p-adically extended integral group ring

Let $\mathfrak{o}$ and $\mathfrak{o}_\mathfrak{p}$ denote the maximal orders of an algebraic number field k and of its $\mathfrak{p}$-adic extension $k_\mathfrak{p}$, respectively. Let G be a finite group of order n. Write (G) for the rational group ring and $[G]$ for the integral group ring. Then $(G)_k$ and $(G)_{k_\mathfrak{p}}$ denote the corresponding group rings with coefficients in k and $k_\mathfrak{p}$; similarly $[G]_\mathfrak{o}$ and $[G]_{\mathfrak{o}_\mathfrak{p}}$ denote the integral group rings after extension of the coefficient ring.

Thus $(G)_k$ and $(G)_{k_\mathfrak{p}}$ are semisimple systems, and $[G]_\mathfrak{o}$ and $[G]_{\mathfrak{o}_\mathfrak{p}}$ are orders in them.

Theorem 1. If the prime ideal $\mathfrak{p}$ does not divide the group order n, then $[G]_{\mathfrak{o}_\mathfrak{p}}$ is a maximal order in $(G)_{k_\mathfrak{p}}$. If $\mathfrak{p}$ divides n, it is not maximal.

The discriminant of the integral group ring, and hence of the extended group rings, is a power of the group order n. If $\mathfrak{p}$ does not divide n, this discriminant is a unit in $\mathfrak{o}_\mathfrak{p}$. Thus $[G]_{\mathfrak{o}_\mathfrak{p}}$ cannot be enlarged to a strictly larger order while keeping the coefficient ring fixed: such an enlargement would split off a non-unit square factor from the discriminant. Since $\mathfrak{o}_\mathfrak{p}$ is already the maximal order of the local field, this proves the first part of the theorem.

If $\mathfrak{p}$ divides n, the direct summand corresponding to the trivial representation gives a genuine enlargement. The idempotent attached to the trivial character has denominator n, and the corresponding order is strictly larger than the integral group ring. Thus the group ring is then not maximal.

Theorem 2. If $\mathfrak{p}$ does not divide n, then every integral or fractional ideal with respect to $[G]_{\mathfrak{o}_\mathfrak{p}}$ is principal.

This follows because $[G]_{\mathfrak{o}_\mathfrak{p}}$ is then a maximal order in a p-adic semisimple system, and ideals in such maximal orders are principal. The assertion is first proved for simple components and then follows for semisimple systems by decomposing into components.

§2. Galois modules, operator-isomorphism, normal bases

Let K/k be Galois with group G. If $S \in G$, write z^S for the conjugate of $z \in K$ under S. The substitutions of G act as operator automorphisms on K/k considered as a k-module. Hence K/k becomes a module over the rational group ring $(G)_k$ by putting

$$z\left(\sum_i c_i S_i\right) = \sum_i c_i z^{S_i}.$$

A module over $(G)_k$ contained in K/k will be called a rational Galois module. Similarly, $[G]_\mathfrak{o}$-modules contained in K/k are called integral Galois modules; in particular $\mathfrak{O}/\mathfrak{o}$ is an integral Galois module.

Theorem 3. As a rational Galois module, K/k is operator-isomorphic to the group ring $(G)_k$.

This is the usual existence of a field normal basis: there is an element $z \in K$ whose conjugates z^S form a k-basis of K. The map

$$\sum_i c_i S_i \longmapsto \sum_i c_i z^{S_i}$$

is an operator homomorphism, and because the two sides have the same rank over k, it is an isomorphism.

Theorem 4. As an integral Galois module, $\mathfrak{O}/\mathfrak{o}$ is operator-isomorphic to a $[G]_\mathfrak{o}$-module of highest rank n inside $(G)_k$. Locally, $\mathfrak{O}_\mathfrak{p}/\mathfrak{o}_\mathfrak{p}$ is operator-isomorphic to a $[G]_{\mathfrak{o}_\mathfrak{p}}$-module of rank n inside $(G)_{k_\mathfrak{p}}$.

Indeed, the rational isomorphism of Theorem 3 carries the integral module $\mathfrak{O}/\mathfrak{o}$ to a module over the integral group ring. Extending the coefficients to $k_\mathfrak{p}$ gives the corresponding local statement.

Theorem 5. At every prime $\mathfrak{p}$ not dividing $n = [K : k]$, the local module $\mathfrak{O}_\mathfrak{p}/\mathfrak{o}_\mathfrak{p}$ has a normal basis.

By Theorem 1, $[G]_{\mathfrak{o}_\mathfrak{p}}$ is a maximal order at such a prime. By Theorem 2, the module corresponding to $\mathfrak{O}_\mathfrak{p}/\mathfrak{o}_\mathfrak{p}$ is a principal ideal:

$$W[G]_{\mathfrak{o}_\mathfrak{p}}.$$

Its basis is WS, where S runs through G. Passing back through the operator-isomorphism, if w is the element corresponding to W, then the conjugates w^S form an $\mathfrak{o}_\mathfrak{p}$-basis of $\mathfrak{O}_\mathfrak{p}$. This is a normal basis.

If $\mathfrak{p}$ divides n, the associated module cannot be principal when no normal basis exists. This is precisely what happens at primes with higher ramification.

The operator-isomorphism of Theorem 3 also gives a concise form of Speiser's results on Klein's problem of forms. Every representation of the Galois group irreducible over k is generated by an irreducible rational Galois module contained in K/k; every absolutely irreducible one is generated after passing to a splitting field Z, using the extended system K_Z/Z. This is exactly the formulation in which a basis $v_1, \ldots, v_m$ of such a Galois module satisfies

$$v_i^S = \sum_j s_{ji} v_j,$$

where $S \mapsto (s_{ji})$ is the corresponding representation.

§3. The discriminant as a group determinant

To decompose the discriminant it is enough to work at the individual ramified primes. In fields without higher ramification, these are precisely the primes at which the normal basis just constructed is available.

Theorem 6. If K/k is a Galois field without higher ramification, then at every ramified prime its discriminant is the square of the group determinant of the Galois group, with the indeterminates replaced by the elements of a normal basis. Corresponding to the irreducible representations, the group determinant decomposes into generalized root numbers; the discriminant decomposes into ideals in the ground field enlarged by the character values, with conjugate-complex characters giving the same ideals.

Let w^S be a normal basis. The discriminant is the square of

$$D = |w^{ST}|_{S,T \in G},$$

which is the group determinant with the w^S in place of the indeterminates. The decomposition of the group determinant according to the absolutely irreducible representations gives factors $D_1, \ldots, D_r$, the generalized root numbers. In the cyclic case these are simply the Lagrange resolvents

$$\sum_S w^S \chi(S).$$

In general, D_i lies in the coefficient field obtained by adjoining the values of the corresponding character. Pairing an irreducible representation with its adjoint gives real factors

$$A_i = D_i \overline{D_i}$$

in the real subfield of the character field. The discriminant ideal is thereby expressed as a product of such factors.

If one can show that the ideals generated by the A_i actually descend to the ground field and are equal for conjugate characters, then they agree with Artin's conductors, provided compound characters are assigned the corresponding products. The functional equation and the behavior for trivial representations of subgroups give the same result for rationally irreducible characters.

Theorem 7. Let $k = \mathbb{Q}$, and let K/k be cyclic of prime degree ℓ, with ℓ prime to the discriminant. Then for the non-trivial representations the ideals generated by the factors A_i are all equal to one another and agree with the conductor of K/k.

This follows from the classical factorization of Lagrange root numbers. At a ramified prime p, after adjoining the ℓ-th roots of unity, the Lagrange resolvents factor in such a way that

$$(A_i) = (D_i \overline{D_i})$$

becomes the same prime-ideal factor for every non-trivial character. Taking norms down to the rational field gives exactly the conductor. The existence of a normal basis, already supplied above, replaces the usual appeal to the realization of absolutely cyclic fields by cyclotomic fields.

38. With R. Brauer and H. Hasse: Proof of a Main Theorem in the Theory of Algebras

We have finally succeeded, by joint effort, in proving the following theorem, which is fundamental for the structure theory of algebras over algebraic number fields and beyond.

Main theorem. Every normal division algebra over an algebraic number field is cyclic, or, in the usual terminology, of Dickson type.

It is a special pleasure to present this result, essentially an achievement of the p-adic method, to Kurt Hensel, the founder of that method, on his seventieth birthday.

The proof consists of three reductions, contributed by the three authors.

1. Hasse's first reduction

The first reduction is due to H. Hasse and is based on his theory of cyclic algebras over algebraic number fields.

Reduction 1. The main theorem follows if the following assertion is proved:

I. Every algebra over Q which is split everywhere is already split over Q.

Here, as in what follows, an "algebra over Q" means a normal simple algebra over the algebraic number field Q, that is, a simple hypercomplex system with center Q. We write

$$A \sim Q$$

when A is a full matrix algebra over Q, i.e. belongs to the special similarity class determined by Q itself as division algebra. An algebra is said to be split everywhere if, for every prime place $\mathfrak{p}$ of Q, the $\mathfrak{p}$-adic extension

$$A_{\mathfrak{p}} = A \otimes_Q Q_{\mathfrak{p}}$$

is split over $Q_{\mathfrak{p}}$.

Let D be a division algebra over Q. Choose a cyclic field Z/Q such that for every prime place $\mathfrak{p}$ of Q the local degree of Z at $\mathfrak{p}$ is a multiple of the local index of D at $\mathfrak{p}$. Then at every prime of Z over $\mathfrak{p}$, the local field $Z_{\mathfrak{P}}$ is a splitting field for $D_{\mathfrak{p}}$. Hence the algebra D_Z obtained by extending the center from Q to Z is split everywhere. If assertion I is known for every ground field, then

$$D_Z \sim Z.$$

Thus D has a cyclic splitting field. Hasse's theory then gives a cyclic splitting field whose degree is the degree of D itself; hence D is cyclic. Therefore the main theorem follows from I.

2. Brauer's second reduction

R. Brauer supplied the decisive idea for the second reduction. He had observed, in the related question of determining the exponent of a division algebra, that Sylow's theorem reduces the problem to the case of a solvable splitting field. The same idea applies to cyclicity.

Reduction 2. Assertion I follows if the following assertion is proved:

II. Every algebra over Q which is split everywhere and has a solvable splitting field is already split over Q.

Let A be split everywhere over Q, and let K/Q be a Galois splitting field for A. Fix a rational prime p, and let L be the fixed field of a Sylow p-subgroup of the Galois group. Then A_L is again split everywhere and has a solvable splitting field, namely K/L. If II is known, then

$$A_L \sim L.$$

Thus A has a splitting field of degree prime to p. The index of A, which divides the degree of any splitting field, is therefore prime to p. Since this holds for every prime p, the index is 1, and A is split. Hence I follows from II.

3. Noether's third reduction

The third reduction, which together with the first two leads to the proof, was found by E. Noether after a communication from Hasse. Hasse had proved the main theorem in the special case of an abelian splitting field by using Reduction 1 and a formula reduction of the corresponding factor system; Noether extracted the essential reduction from that argument. Independently, Brauer had also considered this reduction.

Reduction 3. Assertion II follows if the following assertion is proved:

III. Every algebra over Q which is split everywhere and has a cyclic splitting field of prime degree is already split over Q.

Let A be split everywhere and have a solvable splitting field K/Q. Choose a chain of fields

$$K = A_0 \supset A_1 \supset \cdots \supset A_r = Q$$

such that each A_i/A_{i+1} is cyclic of prime degree. Then A_{A_1} is split everywhere over A_1 and has the cyclic splitting field $K = A_0$ of prime degree. If III is known, A_{A_1} is split. Hence A_1 is a splitting field for A. Repeating the argument down the chain gives finally $A_Q \sim Q$. Thus II follows from III.

But III is known from Hasse's results. Therefore, proceeding backwards through Reductions 3, 2, and 1, the main theorem is proved.

Noether adds that the truth of III, more generally for cyclic splitting fields of arbitrary degree, is equivalent to Hasse's norm theorem. For Reduction 3 only the prime-degree case is required, hence only the original Hilbert-Furtwängler norm theorem. Thus the reduction gives a new simple proof of Hasse's norm theorem.

Indeed, let Z/Q be cyclic and let $\alpha \in Q^*$ have trivial norm residue symbol at every prime place of Q. Then the cyclic algebra

$$A = (\alpha, Z)$$

splits everywhere. By Reduction 3, using only the Hilbert-Furtwängler norm theorem, A is split over Q. But this is equivalent to saying that α is the norm of an element of Z.

The proper generalization of the norm theorem to higher, even non-abelian, cases is therefore not the literal norm statement but assertion I: an algebra which splits everywhere splits globally.

Consequences

The immediate consequence of the main theorem is that Hasse's theory of cyclic algebras over algebraic number fields becomes a general structure and invariant theory of normal simple algebras, in particular normal division algebras, over algebraic number fields.

Theorem 1. The exponent of a normal simple algebra over an algebraic number field is equal to its index.

Theorem 2. The ground ideal of a genuine normal division algebra over Q is divisible by at least one prime place of Q, and in fact by at least two.

Here the ground ideal may be defined as the reduced norm of the reduced different; an infinite prime is included exactly when the algebra reduces there to the quaternion algebra rather than to the real or complex field. Hasse's local theory says that a prime place appears in the ground ideal exactly when the local index is not 1. If all local indices were 1, the algebra would be split everywhere and hence would not be a genuine division algebra. A single exceptional prime is also impossible, because the product formula for the norm residue symbols forces the product of all local invariants to be trivial.

The theorem also gives the arithmetic characterization of splitting fields.

Theorem 3. For an algebra A over Q, an algebraic number field K/Q is a splitting field for A if and only if, for every prime place $\mathfrak{p}$ of Q, the local degree of K at $\mathfrak{p}$ is divisible by the $\mathfrak{p}$-index of A.

Necessity is local. For sufficiency, extend A to K. At each prime of K, the local algebra is split by the divisibility condition. Hence A_K is split everywhere, and by assertion I it is split globally.

Further, the main theorem yields a decomposition theorem for prime divisors in splitting fields and an ordering theorem for splitting fields. If K and K' are splitting fields and $K \subset K'$, then the local degree conditions for K imply those for K'; conversely the partial order among splitting fields is controlled by divisibility of the corresponding local degrees. In particular, minimal splitting fields may be studied arithmetically through local indices.

Finally, the main theorem applies to representations of finite groups. All absolutely irreducible representations of a finite group are realizable over algebraic number fields with the expected splitting behavior, because the simple components of the rational group algebra are normal simple algebras over number fields and hence cyclic algebras. Schur's representation-theoretic indices are therefore exactly the indices of cyclic algebras and are determined by the local invariants.

39. Hypercomplex Systems in Their Relations to Commutative Algebra and Number Theory

The theory of hypercomplex systems, or algebras, has developed rapidly in recent years. Only very recently, however, has the significance of this theory for commutative questions become clear. I will describe that significance through two classical problems going back to Gauss: the principal genus theorem and the closely related norm theorem.

These problems have repeatedly changed their form. In Gauss they appear at the conclusion of his theory of binary quadratic forms. Later they play an essential role in characterizing relatively cyclic and abelian number fields through class field theory. Finally, they can be stated as theorems about automorphisms and splitting of algebras. This last formulation at the same time extends the theorems to arbitrary relatively Galois number fields.

The guiding principle is the use of non-commutative theory for commutative problems. One seeks, through the theory of algebras, invariant and simple formulations of known facts, and then those formulations lead to generalizations.

1. Crossed products

Let K/k be a Galois extension of degree n with group G. A crossed product is a simultaneous embedding of K and the group G into an algebra A in such a way that the automorphisms of K become inner automorphisms in A.

Let u_S be symbols corresponding to the group elements $S \in G$. First put

$$A = \sum_{S \in G} u_S K,$$

so that A is a module of rank n over K. The condition that u_S implement the automorphism S is

$$z u_S = u_S z^S$$

for every $z \in K$. Multiplication among the u_S is prescribed by

$$u_S u_T = u_{ST} a_{S,T}, \qquad a_{S,T} \in K^*.$$

Associativity is equivalent to the factor-system law

$$a_{S,T} a_{ST,U} = a_{S,TU}^U a_{T,U}.$$

The resulting algebra is the crossed product of K with G for the factor system $a_{S,T}$.

Changing u_S to

$$v_S = u_S c_S, \qquad c_S \in K^*,$$

changes the factor system to an associated one,

$$a'_{S,T} = a_{S,T} c_S^T c_T c_{ST}^{-1}.$$

Associated factor systems form one class. Likewise, similar algebras are collected into one algebra class. The classes of crossed products with fixed splitting field K form an abelian group under direct product; this is Brauer's group of algebra classes, expressed in the language of factor systems.

2. Cyclic fields and norms

If K/k is cyclic and S is a generator of its group, the crossed product becomes particularly simple. One may choose one element u corresponding to S, and then

$$uz = z^S u, \qquad u^n = a \in k^*.$$

The factor system is determined by the single element a, and changing u to uc replaces a by

$$a\, N_{K/k}(c).$$

Thus cyclic algebra classes with splitting field K are described by the quotient group

$$k^*/N_{K/k}(K^*).$$

A cyclic algebra (a, K) splits if and only if a is the norm of an element of K.

This is the bridge from the norm theorem to the theorem on split algebras. The norm theorem for cyclic fields says, in one formulation, that if a is a local norm at every finite and infinite place, then a is a global norm. In algebraic language, the cyclic algebra (a, K) splits at every place and therefore splits globally.

The general theorem on algebras says: if an algebra over an algebraic number field splits at every place, then it splits globally. The cyclic norm theorem is exactly its cyclic special case. The proof of the general theorem from the cyclic case proceeds by purely algebraic-arithmetic reductions.

One important consequence, drawn by Hasse, is that every normal simple algebra over an algebraic number field is cyclic. This long-conjectured statement was the problem whose solution led to the general formulation.

3. Principal genus theorem

A second consequence of the theorem on split algebras is the principal genus theorem. Its invariant formulation rests on the fact that the defining relations of the crossed product are purely multiplicative. They continue to make sense if K^* is replaced by an abelian group I on which the Galois group operates.

Taking I to be the group of ideals of K, one obtains factor systems of ideals. A class division of ideals induces a class division of the factor systems, usually finer than the original ideal class division. The identity class of factor systems is defined by those $a_{S,T}$ which generate split algebras at all ramified finite and infinite places.

Let G^* denote the extension of the Galois group obtained in this way. The invariant form of the principal genus theorem is this:

If, under the class division just defined, the substitution

$$v_S = u_S(c_S)$$

induces an automorphism of G^*, then this automorphism is inner and is generated by an ideal class (b).

Equivalently, if the transformation factors

$$(c_S)^T(c_T)(c_{ST})^{-1}$$

belong to the identity class of factor systems, then there is an ideal class (b) such that

$$(c_S) = \frac{(b)}{(b)^S}$$

for all $S \in G$. In the cyclic case this says that if the norm of an ideal class lies in the identity class of factor systems, then the class is a symbolic $(1-S)$-th power.

4. Relation with the classical form

If one restricts to ideals prime to the ramified places, the content remains the same. For quadratic fields the principal genus just defined agrees with Gauss's principal genus. Ideal classes correspond to quadratic forms, and norms of ideal classes correspond to integers represented by those forms. The condition that the identity class generate split algebras at the ramified places says exactly that those represented integers are quadratic residues at the ramified places. Thus the forms have the total character of the principal form.

For cyclic fields the theorem becomes the usual statement: the principal genus consists of those ideal classes whose norms are norm residues at the finite and infinite ramified places. This is equivalent to being a norm residue modulo the conductor.

In the general Galois case one can likewise introduce a conductor built only from the ramified primes, normalized so that the factor systems at each ramified place contain only elements of the identity class. The relation with Artin's conductors, which involve the same prime ideals, leads back to the theory of Galois modules and hence to the second hypercomplex method. How far these two methods reach remains a question for future work.

Non-commutative Algebras

It is well known that the main theorems of commutative algebra are contained in Galois theory, preceded by the theory of detachment fields and splitting fields: fields large enough, respectively, to split off one linear factor of a given polynomial or to decompose it completely into linear factors.

The corresponding parts of algebra are developed here in the non-commutative setting, and especially for hypercomplex systems. The method is deliberately non-commutative: it works with representations in non-commutative division rings. At the end I show how the corresponding theorems of commutative algebra may be proved in a completely parallel manner by representation in commutative fields.

The representation theory used here is a further development of the theory based on representation modules. It is preceded by a short automorphism theory. Alongside direct representations I consider reciprocal representations. Each can be reduced to the other, but the reciprocal representation is generated by the reciprocal representation module. The advantage is that this reciprocal representation module is a bimodule and may also be regarded as a one-sided module over an extension ring. Thus representation in non-commutative division rings is reduced to ideal theory in the extension ring: the irreducible representation classes correspond to the irreducible ideal classes of that extension ring, in exact analogy with the facts for representations of hypercomplex systems over their commutative coefficient domain.

From this one obtains structure theorems for matrix rings over non-commutative division rings. The basic observation is that every subring defines a representation by the division ring, or reciprocally by the oppositely isomorphic division ring. This oppositely isomorphic division ring is the first analogue of a minimal detachment or splitting field in the commutative case: it mediates all reciprocal representations of first degree and, when the division ring has finite rank over its center, also gives a complete splitting into direct summands of rank one.

This yields the Galois theory for division rings and at the same time expresses the fact that reciprocally isomorphic division algebras generate inverse classes in Brauer's group of algebra classes. A second proof of the Galois theory, applying more generally to simple systems, is almost an immediate consequence of a theorem on mutually commuting subrings. That theorem itself is tied directly to the preliminary automorphism considerations.

Up to this point the arguments are purely non-commutative. But the problem of commutative splitting fields is also handled non-commutatively, by representing the splitting field in the division algebra. The final sections transfer the same proof method to the commutative case and develop splitting fields for arbitrary systems, thereby giving the connection with the ordinary commutative representation theory.

§1. Automorphisms, modules, and bimodules

The representation theory based on representation modules rests on the theory of the automorphism ring of abelian groups with or without operators. The elementary arguments concerning mappings and laws of calculation will be collected here to avoid repetition.

1. Multiplicative mapping and associativity

Let G first be a group without operators, and let A be its absolute endomorphism domain, the set of all homomorphisms of G into itself. This domain is multiplicatively closed. If $s, t \in A$, the product st is defined by

$$g(st) = (gs)t,$$

and associativity is immediate.

Now let G be a group with operators. Thus there is a set B of symbols $O, H, \ldots$ such that $gO, gH, \ldots$ are uniquely defined elements of G and each symbol induces an endomorphism of G. Hence there is a definite, though not necessarily one-to-one, map from B into the absolute endomorphism domain A.

If the operator domain B is itself multiplicatively closed, then the map from B to A is multiplicatively homomorphic if and only if the corresponding associativity relation

$$g(OH) = (gO)H$$

holds for all $g \in G$ and $O, H \in B$. For left operators the same assertion gives a reciprocal homomorphism, since left multiplication must be read in the opposite order whereas operators are read from left to right.

2. Operator-homomorphic mapping

If G is a group with operator domain B, a mapping $G \to G'$ is operator-homomorphic when it is a group homomorphism and respects the operators. When the operator domains themselves are also mapped, this includes ring homomorphisms as a special case.

The essential principle is this. Whenever a group is made into a group with operators by a multiplicatively closed operator domain, the associativity law is exactly the statement that the operator domain maps homomorphically into the endomorphism ring. Conversely, a homomorphic image of an operator domain in endomorphisms supplies the required associative action.

3. Modules and bimodules over rings

Let R be a ring and M an abelian group. A right R-module structure on M is an operator structure satisfying

$$(m_1 + m_2)r = m_1 r + m_2 r,$$

$$m(r_1 + r_2) = mr_1 + mr_2,$$

$$m(r_1 r_2) = (mr_1)r_2.$$

Left modules are defined similarly, with the order in the last formula reversed. If M is a right module over R and a left module over S, and

$$s(mr) = (sm)r,$$

then M is a bimodule.

The automorphism ring of a module is the ring of all module endomorphisms. If these endomorphisms are written on the side opposite the original ring action, a module becomes a bimodule over the original ring and its automorphism ring. Conversely, every bimodule determines homomorphisms from one of its multiplier rings into the automorphism ring relative to the other.

4. Passage from a right bimodule to a one-sided module

Suppose M is a right module over a ring R, and suppose that R contains two subrings which commute elementwise. Then M can be regarded as a one-sided module over the extension ring generated by those two subrings. Conversely, a one-sided module over such an extension ring gives a bimodule by restriction to the two commuting subrings.

This elementary passage is the key technical device. It permits one to replace reciprocal representation modules, which are naturally bimodules, by one-sided modules over an extension ring. Ideal theory for the extension ring can then be applied.

§2. Representation modules

1. Modules of linear forms

The transition from the preceding automorphism theory to representation theory proceeds through modules of linear forms. Let a hypercomplex system S act on a module M by right multiplication. Choosing a basis of M over the coefficient division ring gives matrices for the action of every element of S. The entries of those matrices depend on the chosen basis, but the module class does not.

The automorphism ring of a module of linear forms commutes with the representing matrices. Conversely, matrices commuting with a given representation arise as endomorphisms of the corresponding representation module. This is the module-theoretic form of Schur's lemma and its generalizations.

Theorem. If M is a module of linear forms for a system S, then its S-automorphism ring is the complete commutant of the matrices representing S on M.

In particular, when M is irreducible, this automorphism ring is a division ring.

2. Representation and representation module

A direct representation of a system S assigns to every element $s \in S$ a matrix in such a way that sums and products are respected. The representation module is the module whose basis vectors are the rows or columns on which these matrices act. Equivalent representations yield operator-isomorphic modules, and operator-isomorphic representation modules yield equivalent representations.

A reciprocal representation is defined by reversing the multiplication order. It is generated by the reciprocal representation module. The reciprocal module is naturally a bimodule: the represented system operates on one side, and the coefficient division ring operates on the other.

Theorem. Every reciprocal or direct representation module generates a representation, and equivalent modules generate equivalent representations. Conversely, a representation determines its module class.

The usefulness of the reciprocal module is that it may be viewed as a one-sided module over a suitable extension ring. Thus questions about irreducible reciprocal representations become questions about simple one-sided ideals in that extension ring.

3. Passage from reciprocal modules to ordinary modules

Let A be the coefficient division ring and let S be represented reciprocally in A_n, the full matrix ring over A. The reciprocal representation module is a right module over an extension ring generated by S and A. By the preceding passage, it may be treated as a one-sided module.

The consequence is the following fundamental connection. The classes of irreducible reciprocal representations of S over A correspond one-to-one to the irreducible ideal classes of the extension ring. Under complete reducibility, direct-sum decompositions of the module correspond to decompositions of representations into irreducible components.

This is the non-commutative analogue of the familiar fact that, for a hypercomplex system represented over its own commutative coefficient field, irreducible representations are obtained from one-sided ideals.

4. The theorem on commuting matrices

The module formulation gives the theorem on commuting matrices. If a system of matrices over a division ring is irreducible, its commutant is a division ring. If the representation is completely reducible, the commutant is a full matrix ring over the division ring attached to the irreducible class, with one full matrix factor for each multiplicity.

Conversely, if two systems of matrices are full mutual commutants in a full matrix ring, each is completely reducible relative to the other. This double-centralizer idea is the mechanism behind the later theorem on mutually commuting subrings.

§3. Extension modules

1. Normal bases of submodules

Let A be a division ring and let P be a subfield of its center or, more generally, a central subfield. If M is a module over A, a P-submodule generated by a basis may be extended by multiplication with A to the whole module. Linear independence over P and over A are related exactly as in the commutative case once left and right conventions are fixed.

This gives a normal-basis type statement for modules. A suitable basis of a submodule may be completed to a basis of the larger module after scalar extension. Such bases are used to compare ranks in the double-centralizer calculations.

2. Extension modules

Let A be a division ring and P a subfield. An A-module M of rank n is called an extension module of P-rank r when it is generated over A by a P-module of the corresponding rank and when the two scalar structures are compatible. The extension module records precisely how a representation over a smaller field becomes a representation over the larger division ring.

If $z_1, \ldots, z_m$ are linearly independent over P, then their images remain independent under suitable scalar extension. Every P-module

$$T = z_1P + \cdots + z_mP$$

inside M generates an extension module by multiplication with A.

3. Invariant modules

Let $M = M_1 + \cdots + M_r$ be a direct decomposition into simple components. A submodule invariant under a second commuting system corresponds to a sum of certain components. The precise theorem is that extension modules of invariant submodules and invariant submodules of extension modules are in one-to-one correspondence.

The proof is a straightforward use of the uniqueness of direct-sum decompositions into simple modules. If a submodule is invariant, its projection onto each simple summand is either zero or the full summand. Conversely, any such sum is invariant.

This will be used to identify two-sided ideals with invariant modules.

§4. Simple systems

1. Simple systems and modules

Let S be a simple hypercomplex system over P. A representation over a division ring A is irreducible if its representation module is simple. The corresponding extension ring is then a simple ring. Conversely, a simple ideal class of the extension ring gives an irreducible representation.

If S_A denotes the extension of S to A, then a decomposition of S_A as a module reflects the decomposition of the representation after extension. When A is large enough to split the system, the simple components have rank one in the relevant sense, and the representation is absolutely irreducible.

2. Representation classes

A representation class is a class of operator-isomorphic representation modules. For a simple system with center P, representation classes over a division ring A correspond to simple ideal classes in the extension ring. The number of irreducible classes, their dimensions, and their automorphism division rings are obtained from the decomposition of this extension ring.

Theorem on representation classes. Let S be hypercomplex over P and let A be a division ring containing P in its center in the appropriate sense. The irreducible reciprocal representation classes of S in A are in one-to-one correspondence with the irreducible right ideal classes of the extended system. The direct-sum decomposition of the extended system gives the decomposition of any completely reducible representation.

3. Rank relation

When S is simple, the rank of an irreducible representation, the rank of the corresponding ideal, and the rank of the associated division algebra satisfy the usual product formula. If a simple algebra S is embedded irreducibly in a full matrix ring over A, and if B is the commuting division ring, then the product of the ranks of S and B equals the rank of the full matrix ring. This is the numerical form of the double-centralizer theorem.

4. Sharpened rank relation

If A has finite rank over its center, the rank relation can be sharpened. The division ring reciprocal to A appears as the full commutant of an irreducible embedding. The two central simple algebras obtained in this way have reciprocal classes in the Brauer group. Their indices multiply in the expected manner, and splitting of one corresponds to complete reduction of the other.

§5. Commuting subrings

1. Reducible and irreducible embeddings

Let A_n be a full matrix ring over a division ring A, and let S be a subring. An embedding of S is irreducible if the natural module has no proper invariant submodule. It is completely reducible if the module is a direct sum of irreducible invariant submodules.

If the embedding is reducible, the matrices of S may be simultaneously put into block triangular form. If it is completely reducible, they may be put into block diagonal form. These statements are merely the module-theoretic reduction theorems translated into matrices.

2. Inner automorphisms

If two simple subrings of A_n both contain the center P and are isomorphic over P, then under suitable irreducibility hypotheses an isomorphism between them is induced by an inner automorphism of A_n. That is, there exists an invertible element $u \in A_n$ such that

$$\varphi(s) = u^{-1}su.$$

This is the non-commutative analogue of the extension of field isomorphisms.

The proof uses the representation modules attached to the two embeddings. Since the embeddings are irreducible and isomorphic, the modules belong to the same class. An operator-isomorphism of the modules is given by multiplication with an invertible matrix, and this matrix implements the desired inner automorphism.

3. The theorem on mutually commuting subrings

The central theorem is the following.

Theorem. Let S and T be simple subrings of a full matrix ring over a division ring, each containing the center P, and suppose that every element of S commutes with every element of T. Then the full commutant of S is generated by T together with the center in the appropriate way precisely when the corresponding representation is irreducible; in general the two full commutants are obtained by passing to the completely reducible decomposition.

In the finite-rank case this becomes the usual double-centralizer theorem: if S is a central simple subalgebra of a central simple algebra A, then the centralizer $C_A(S)$ is central simple over the same center, and

$$[S:P][C_A(S):P] = [A:P].$$

Moreover, S and its centralizer are mutual centralizers.

4. Sharpened commutation theorem

If S is a simple subring and A is a division algebra finite over its center, then the centralizer of S in a full matrix algebra over A has class inverse to the class represented by the embedding of S. In particular, if one passes from S to its reciprocal system, the associated algebra class is inverted in the Brauer group.

This sharpened theorem is the bridge to Galois theory. It says that intermediate division rings and closed subgroups of the inner automorphism group determine one another through centralizers.

§6. Galois theory of simple systems

The sharpened commutation theorem expresses the Galois theory of simple systems relative to the center. We first consider division rings, then simple systems in general.

1. Galois theory of division rings finite over the center

Let A be a division ring with center P. Its Galois group is the group of automorphisms of A which extend the identity on P. By the Skolem-Noether theorem in this setting, these are the inner automorphisms. Thus the group is isomorphic to

$$A^*/P^*,$$

where A^* and P^* denote the multiplicative groups of non-zero elements.

Subgroups of this group correspond to subgroups H^* of A^* containing P^*. A subgroup is called closed if, after adjoining zero, H^* becomes a division subring H of A.

The statement that a division subring C is fixed by a subgroup is equivalent to saying that C commutes elementwise with the division ring H attached to the subgroup. Therefore the commutation theorem gives the main theorem of non-commutative Galois theory:

Main theorem of the Galois theory of division rings. The division rings C between P and A and the closed subgroups H of the Galois group correspond one-to-one, in such a way that C is the full invariant division ring of H and H is the full invariant group of C.

2. Galois theory of simple systems

Let A now be a simple system with center P. Its Galois group is again the group of inner automorphisms, isomorphic to A^*/P^*, where now A^* denotes the group of regular elements of A.

A subgroup is called simply closed if the subring generated by the corresponding regular elements is a simple system and if the group consists of all regular elements of that simple system, modulo P^*. The key lemma is that every simple system containing P is generated by its regular elements. This is clear when P is infinite, since a general element with indeterminate coefficients is regular and may be specialized to regular basis elements. Shoda proved it in general.

Using this lemma, commutativity with a simple system is equivalent to invariance under the automorphisms induced by its regular elements. Hence the commutation theorem gives:

Main theorem of the Galois theory of simple systems. The simple systems S in A containing P and the simply closed subgroups of the Galois group correspond one-to-one, in such a way that S is the full invariant domain of the subgroup and the subgroup is the full invariant group of S.

3. Proof by the extension principle

For division rings there is another proof of the main theorem, based on extending isomorphisms, i.e. representations of first degree. It avoids rank counting and therefore requires fewer finiteness hypotheses. It also shows in what direction an extension to division rings of infinite rank might be possible.

Assume that a simple system S over P admits a reciprocal representation of first degree in A, so that S is itself a division ring. Let

$$S_A = e_1 S_A + \cdots + e_n S_A$$

be a decomposition into simple right ideals. The representation generated by e_i is defined by $e_i w = e_i w$ in A.

Lemma 1. The n reciprocal representations generated by $e_1, \ldots, e_n$ are distinct. Thus S has at least as many distinct representations as its rank.

If two of them were the same, then $e_i w = e_j w$ for every $w \in S_A$. But this is impossible because $e_i e_j = 0$ for $i \neq j$.

Lemma 2. Let T be an intermediate division ring between P and S, and let $s = [S : T]$. Then every reciprocal isomorphism of T into A has at least s distinct extensions to S.

Indeed, decompose the representation of T by idempotents. Multiplication by S splits each component into exactly s components, and Lemma 1 gives their distinctness.

Define the class division of the reciprocal isomorphisms of S induced by T: two isomorphisms are equivalent if they induce the same isomorphism on T. Then:

Main theorem in this form. If $\mathfrak{I}_T$ is the class division induced by T, then T is a maximal intermediate division ring with respect to this class division.

Passing from the set of isomorphisms to the automorphism group by multiplication with the inverse of one fixed isomorphism gives the usual subgroup formulation. The assertion that closed subgroups are full invariant groups is included by taking the division ring H attached to such a subgroup as T.

§7. Similarity classes of algebras and splitting fields

From now on A, B, and their matrix rings are assumed finite over their centers P. Thus they are normal simple algebras over P; A and B may be division algebras.

All algebras A_n with the same associated division algebra A are collected into one similarity class. Isomorphic division algebras are identified.

1. The group of algebra classes

If A and B are algebras over P, then their direct product over P is again an algebra over P. At the level of associated division algebras this product determines a class, independent of matrix factors. Thus the classes form a multiplicatively closed commutative system.

Brauer's theorem states that these similarity classes form an abelian group under direct product. The identity is the class of full matrix algebras over P. The inverse of the class of A is represented by the algebra $\bar{A}$ reciprocally isomorphic to A. The relation

$$A \otimes_P \bar{A} \sim P$$

follows from the sharpened commutation theorem.

This is the Brauer group of P.

2. Splitting fields

An extension field Z/P is a splitting field of A if

$$A_Z \sim Z.$$

Equivalently, Z contains a maximal commutative subfield of the division algebra associated with A, or at least is large enough to make the associated division algebra into a matrix ring.

A field is a detachment field if it splits off one simple component of the representation; in the central simple case detachment and splitting agree for maximal commutative subfields. In general, every maximal commutative subfield of a division algebra is a splitting field, and every splitting field has degree divisible by the index.

3. Rank relations and index

Let A be central simple over P, and let Z be a maximal commutative subfield. If $[A : P] = m^2$, then $[Z : P] = m$. The integer m is the Schur index. It is the degree of any maximal commutative subfield embedded in the division algebra, and it is also the minimal possible degree of a splitting field in the regular case.

For a general splitting field of degree n, one has

$$n = mr$$

for some integer r. This follows either from rank comparison or from the fact that the index divides every splitting-field degree.

If A is embedded in a larger division algebra D and L is the centralizer, then the refined relation reads

$$[A : P][L : P] = [D : P]$$

with the appropriate square-root conventions for indices. These are exactly the double-centralizer rank formulas.

4. Existence of separable splitting fields

Every algebra class has separable splitting fields, even such as can be embedded in the algebra itself. Let the ground field have characteristic p, and write the index as

$$m = sp^r,$$

with s prime to p. Suppose first that a splitting field has inseparable part of degree p^r. One shows that this inseparable part can be reduced. The reduced discriminant of the division algebra after extension to an algebraic closure is non-zero, so there exists an element with non-zero reduced trace. Such an element cannot lie in the purely inseparable ground part, since traces there vanish by multiplication by p. Adjoining it gives a proper separable extension and reduces the remaining index. Repetition yields a separable splitting field of degree equal to the index.

§8. Splitting fields, detachment fields, and Galois theory in the commutative case

We now pass from splitting fields of central simple systems to those of arbitrary simple systems by developing the splitting theory of the center. Everything in this paragraph is commutative.

1. Commutative simple systems

A commutative simple system Z over P is a field extension. Let Ω be an algebraically closed extension field of P. Then Z_Ω has no radical if and only if Z/P is separable. Equivalently, Z has as many distinct embeddings into Ω as its degree.

An extension field A of P is called a splitting field of Z if Z_A decomposes into composition factors of first degree. It is called a detachment field if at least one composition factor of first degree splits off. Thus A is a detachment field if it contains a subfield isomorphic to Z, and is a splitting field if it contains a field isomorphic to the Galois closure of Z.

A minimal detachment field is isomorphic to Z. It is also a splitting field if and only if Z/P is normal.

2. Galois theory for separable fields by the hypercomplex method

Assume Z/P is finite separable, and let Ω be a splitting field, finite or infinite. Then

$$Z_\Omega = e_1 Z_\Omega + \cdots + e_n Z_\Omega$$

decomposes into a direct sum of fields corresponding to the distinct embeddings of Z into Ω. The idempotents e_i determine the embeddings.

If T is an intermediate field between P and Z, and $s = [Z : T]$, then every embedding of T into Ω has exactly s extensions to Z. This is proved exactly as in the non-commutative extension-principle proof: the decomposition of T_Ω refines to that of Z_Ω, and the number of idempotents above each one is s.

The class division of the embeddings of Z induced by T has T as its maximal intermediate field. When Z/P is Galois, composition with the inverse of a fixed embedding identifies the embedding set with the automorphism group, and one obtains the usual Galois correspondence.

3. Idempotents of Z_Ω

Let S run through the embeddings which carry Z into its conjugate copies inside the splitting field. The idempotents e^S of Z_Ω are conjugate:

$$e^S = eS,$$

with $e = e^1$. Applying S to the defining relations of e carries them to the defining relations of e^S. Since the direct-sum decomposition is unique, the idempotents are uniquely determined and hence conjugate.

If Z/P is Galois, this immediately gives the subgroup part of the main theorem of Galois theory. For if H is a subgroup, the sum

$$\sum_{S \in H} e^S$$

is fixed by all substitutions in H and by no larger subgroup. Hence the invariant field attached to H is recovered.

4. Complementary bases

Let $a_1, \ldots, a_n$ be a basis of Z/P. Write the idempotent in the form

$$e = a_1\beta_1 + \cdots + a_n\beta_n.$$

Then, for every embedding S, the element

$$e^S = a_1\beta_1^S + \cdots + a_n\beta_n^S$$

expresses the image of the basis complementary to $a_1, \ldots, a_n$ in the embedding defined by e^S. The relations

$$e^S e^R = 0 \quad (S \neq R), \qquad (e^S)^2 = e^S$$

translate precisely into the defining equations for complementary bases. Thus the contragredience of complementary bases follows from the invariance of the idempotents, independently of the chosen basis.

This recovers Dedekind's relation between complementary bases and idempotents in the language of hypercomplex decompositions.

§9. Splitting and detachment fields of arbitrary systems

1. Definitions and reduction to the simple case

Let S be hypercomplex over P. A commutative extension field A/P is a splitting field of S if, in S_A, every composition series by one-sided ideals has absolutely simple composition factors. Equivalently, all irreducible representations of S over A are absolutely irreducible. An extension field T/P is a detachment field if S_T splits off at least one absolutely simple composition factor, equivalently if at least one irreducible representation over T is already absolutely irreducible.

These definitions include the earlier ones: for commutative simple systems they reduce to the definitions of §8, and for central simple algebras they reduce to the definitions of §7. In the latter case detachment and splitting fields coincide, because after a commutative extension of the center the algebra remains completely reducible, and all simple components are operator-isomorphic. Splitting off one absolutely simple component forces complete splitting.

The splitting fields of a general system are obtained by taking the compositum of splitting fields of the irreducible representations over P. A detachment field of the system is a detachment field of at least one irreducible representation. Because irreducible representations correspond one-to-one to the simple components of the radical quotient, it is enough to consider simple systems.

2. Simple systems with separable center

Let A be a simple system over P with center Z, and assume Z/P is separable. Let Ω be a splitting field of Z. The direct-sum decomposition of Z_Ω induces a two-sided decomposition of A_Ω:

$$A_\Omega = e_1 A_\Omega + \cdots + e_n A_\Omega.$$

The components are conjugate and are isomorphic to A in the appropriate sense; each is a normal simple algebra over its own center $e_i Z_\Omega$.

Therefore A_Q has no radical for every extension field Q of P; it is completely reducible. The components remain simple after arbitrary coefficient extension, because the center has already separated. The detachment and splitting fields are then characterized by combining §7 with §8.

If A is a division algebra with separable center Z, the irreducible embeddings of detachment fields are precisely all maximal commutative subfields of A which contain a copy of Z. Splitting fields are precisely the composita of a detachment field with a splitting field of the center, in particular with the Galois closure of Z.

A minimal detachment field can be a splitting field even if the center is not Galois. This is one of the differences from the commutative case: for example, a quaternion algebra with center isomorphic to a quadratic field may have the Galois closure of the center already as a minimal detachment and splitting field.

3. Inseparable center

If the center is inseparable over P, then Z_Ω has a radical, and consequently A_Ω has a radical. Let $\mathfrak{C}$ be the radical of Z_Ω, and let $\mathfrak{c}$ be the ideal it generates in A_Ω. The quotient $Z_\Omega/\mathfrak{C}$ decomposes into conjugate components, exactly as in the separable case, and this induces a direct-sum decomposition of

$$A_\Omega/\mathfrak{c}.$$

The components are again isomorphic to the original simple system in the corresponding sense and have centers equal to the corresponding components of $Z_\Omega/\mathfrak{C}$.

Rank counting shows that the composition factors belonging to a composition series of $\mathfrak{c}$ are operator-isomorphic to those of $A_\Omega/\mathfrak{c}$, not merely homomorphic. Hence the representation-theoretic conclusions remain valid after replacing direct-sum components by composition factors.

4. Final representation-theoretic formulation

For irreducible representations the result may be summarized as follows.

Let a representation of a hypercomplex system be irreducible over P. It becomes absolutely completely reducible after extension of the coefficient field if and only if the corresponding center is separable over P. In every case, splitting fields and detachment fields of the representation can be characterized by embeddings in the associated simple system as above.

In particular, the lowest possible degree of a detachment field is the product of the degree of the center over P and the index of the corresponding algebra class over that center. The degree of every detachment field is a multiple of this number.

The representation splits, in the sense of composition series, into as many distinct conjugate absolutely irreducible representation classes as the degree of the largest separable subfield contained in the center. Each class occurs with multiplicity equal to the product of the index and the degree of the center over that separable subfield.

When the ground field is perfect, so that all finite extensions are separable, this recovers the familiar results of I. Schur, with the additional embedding characterization already found in Brauer's work.